Level Sensor Deconvolution

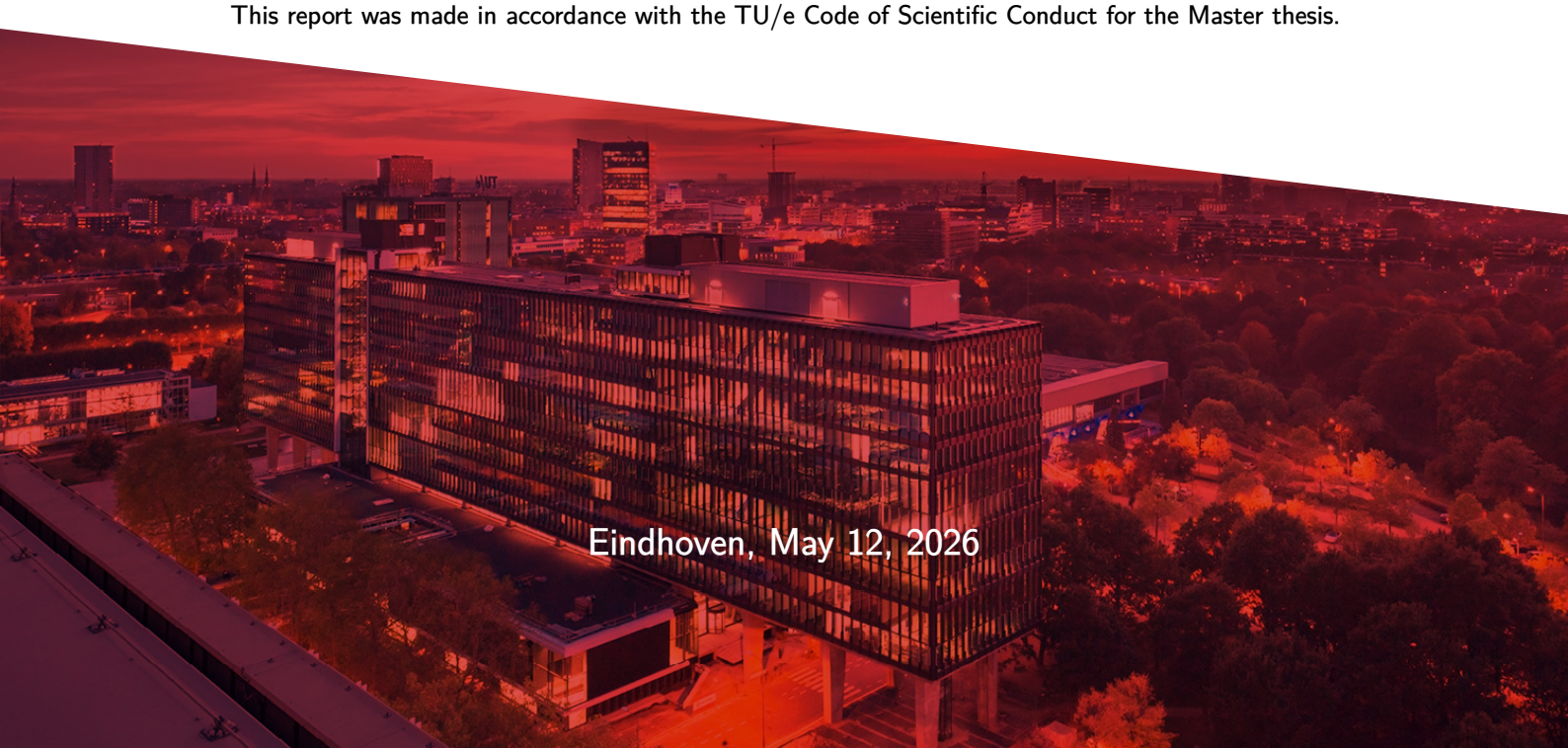
Master Thesis Industrial and Applied Mathematics

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1 | Introduction

ASML develops *lithography* machines that are essential for the production of integrated circuits. In these lithography machines, patterns are repeatedly printed onto a layer of light-sensitive material called *photoresist*. This photoresist layer has been deposited on a silicon wafer before this, and the pattern is printed onto the photoresist using a light source. The printing of a pattern onto a layer of photoresist on a wafer is also referred to as *wafer exposure*. A wafer is a very thin round slice of silicon on which layers of resist are deposited, exposed, and treated. The photoresist is chemically treated after each round of exposure to create patterns, enabling the construction of electronic components with nanometer precision. In the process, it is crucial to have a very accurate estimate of the height profile of a wafer before the wafer gets exposed.

During the exposure process, for the projection lens to be in focus, the wafer has to be in the right range of height. See the image on the left in Figure 1.1 for a visualization of this. Naturally, we need to know the height of the wafer in the area that we expose to ensure the surface of the wafer is in this range. Therefore, before the exposure process, we have a measure process in which the *level sensor* measures the height of the wafer. The mechanism through which the level sensor does this is schematically represented in the image on the right in Figure 1.1.

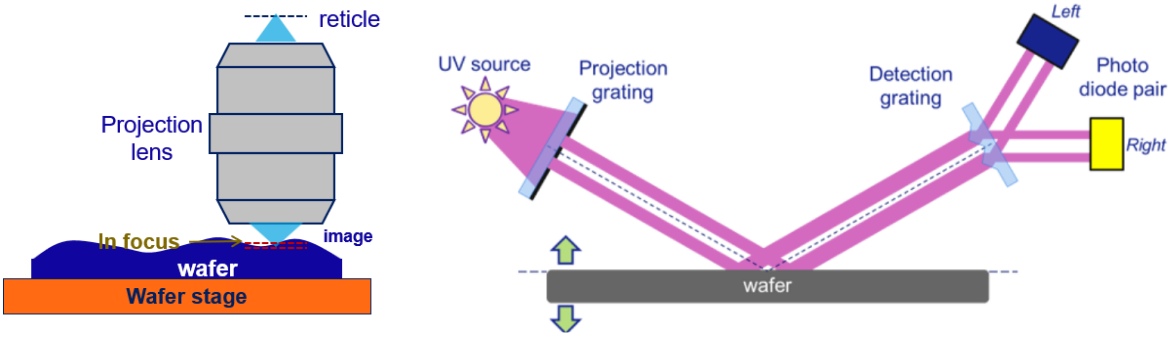


Figure 1.1: (Left) Schematic representation of the projection lens and focus range. (Right) Schematic representation of the level sensor measuring the height of the wafer.

The level sensor has a number of *spots* that are aligned in the x-direction. Light is shone through a projection grating. The light then hits the wafer in these spots, the wafer surface reflects this light and the light goes to the detection part of the sensor. In this detection part, there are prisms that split the light into a left (L) and a right (R) receptor. The ratio $\frac{R-L}{R+L}$ tells us something about the height of the wafer: if the wafer is lower, the reflected light will consequently enter the prisms in the detection part of the sensor from a lower angle resulting in different amounts of L and R.

The process that we described is how the level sensor performs a single measurement on the wafer, but of course, we need many such measurements to get an overview of the entire height profile of the wafer. For this reason the level sensor performs *strokes* over the wafer. A single stroke is synonymous with the level sensor scanning over the entire length of the wafer in the y-direction and making many of these measurements along the way. The y-direction (including plus and minus) is also called the *scan direction* for this reason. Keep in mind that the wafer is too wide for a single stroke to cover the entire wafer surface, so multiple strokes are necessary to cover the entire wafer surface. This process is visualized in Figure 1.2.

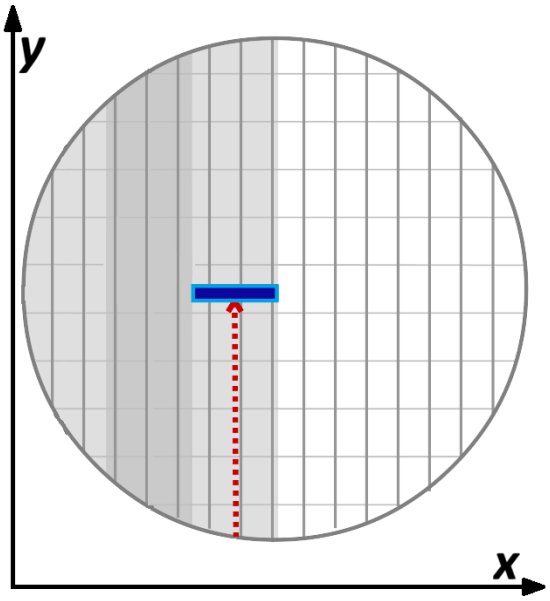


Figure 1.2: The level sensor scans the surface of the wafer in so-called *strokes*.

Of course, these measurements of the height of the wafer done by the level sensor are not perfectly accurate; there are measurement artifacts and multiple sources of noise present. Moreover, the scale of the features on the wafer is smaller than the size of the spots used to measure the wafer. So the measurements we receive from the level sensor also include a convolution in the form of a moving average of the height profile of the wafer. This is exactly where the main research question of this project comes in: how can we, given the convolved and noisy measurements from the level sensor, retrieve the actual height profile and topography of the wafer? In the next section, we describe this *inverse problem* in more detail.

1.1 | The inverse problem

The central problem of this thesis is the following inverse problem. We observe measurements from the level sensor, and from these measurements we want to recover the actual height profile of the wafer before the signal was convolved and filtered by the machine.

More specifically, let

$$h(x, y) : \mathbb{R}^2 \to \mathbb{R}$$

denote the true wafer height profile. The level sensor does not give us this height profile directly. Instead, the measured signal is affected by multiple factors.

Firstly, the measured signal is affected by the level sensor's moving average kernel, since the height values within a spot are averaged over. Secondly, the measured signal is affected by the *curved wafer effect* (CWE), which is a physical phenomenon that occurs when the light deflects at sharp height steps. Moreover, in the case of filtered data, the signal is also impacted by the *driver filter*, which is an additional machine-side filtering step that is applied to the raw level sensor signal. Of course, there is also noise present in the signal, so what we observe is a transformed version of the original wafer height. In the internship report [5], we worked with simulated raw level sensor data, so this is data before application of the driver filter. Moreover, the model for the level sensor only included an averaging kernel and excluded the CWE.

In general terms, the problem is therefore:

> given measured level sensor data z_{meas} with the distortions mentioned above,
> recover the original wafer height profile h.

This is exactly why we are dealing with an inverse problem: we observe the output of a physical measuring process and want to infer the input that produced it. There are two main difficulties with this. Firstly,

the measuring process itself smooths the signal: small features of the wafer are averaged by the level sensor spot and the curved wafer effect introduces additional distortions near sharp height steps. Secondly, the final signal that is most available in practice is the filtered data, that has been impacted by the driver filter. So the measured data does not contain the original sharp profile directly, but only an indirect and distorted version of it.

In this thesis, we focus on a class of wafers for which the field layout is known a priori. We also call this known layout the *nominal* field layout. A field is a section on the wafer, and the wafer is divided into many of these fields. For visualization, a field can be thought of as one of the rectangles on the wafer in Figure 1.2. The field has a certain pattern called the field layout, which is recurring on the wafer.

The field layout for this class of wafers can be described as a step profile or piecewise constant profile in the y-direction. Knowing the field layout means that we know where these step locations in the scan direction are. Because of this, the inverse problem becomes a parameter-estimation problem: instead of trying to recover an arbitrary function $h(x, y)$, we may assume a piecewise constant profile with known segment boundaries, which are the step locations, and we then solve for the unknown segment heights.

This thesis considers two inverse problems. In the first, the forward model is assumed to be fully known and hence kept fixed, and only the segment height parameters are unknown. In the second, part of the level sensor kernel is also treated as unknown, so that we simultaneously estimate the wafer height profile and a kernel parameter. The precise mathematical formulation of these two problems will be given in Section 2.1, and the corresponding algorithms will be described in Section 3.1.

In the internship report [5], we investigated a problem similar to the first inverse problem as described above, but for simulated raw level sensor data, which was approximately 10 times more dense than the real filtered data that we focus on in this thesis. Moreover, the level sensor model used was simpler and less accurate.

1.2 | Results

The main goal of this thesis is to formulate and implement a deconvolution method that is able to retrieve the original step profile of the wafer with good accuracy from level sensor measurements. Additionally, in the second inverse problem, our goal is to formulate and implement a deconvolution method that is able to retrieve the original step profile and an additional kernel parameter simultaneously with good accuracy from level sensor measurements. In particular, the emphasis lies on methods that are compatible with the real, filtered data that is most available in practice.

Our first result is that a least-squares approach based on a piecewise constant model for the wafer height profile performs very well on synthetic data. Using an ASML simulation tool for the level sensor kernel and for the driver filter, synthetic measurements that include realistic noise are generated on a 0.2 mm grid. When the nominal segment boundaries are assumed known, the least-squares method is able to recover the original alternating step profile with very high accuracy. This serves as an important validation of the method.

The second result is that the same least-squares framework can also be applied to real, filtered level sensor data. In the case of real data, an additional interpolation step is required. The method yields reconstructed profiles that, after evaluation, match the measured data well and recover the characteristic step structure with segment height parameters that are very close to what we know to be true.

The third main result is that a first semi-blind deconvolution experiment is feasible. More specifically, a kernel parameter τ for the level sensor related to the curved wafer effect is treated as unknown and estimated together with the segment heights by means of *variable projection*. In the synthetic setting, this method is able to recover the true kernel parameter accurately and at the same time still reconstruct the underlying height profile very well. This shows that it can be meaningful and useful to move beyond a completely fixed forward model and to estimate part of the kernel from the data as well.

The fourth main result is that the one-dimensional real data reconstruction can be extended to a two-dimensional map by repeating the one-dimensional reconstruction independently for all spots in a

stroke. Note however that this does not yet constitute a full two-dimensional deconvolution problem in both x and y, but it does yield a spatial map of the reconstructed field at the discrete spot locations in the x-direction. In this way, the recovered step profile can be visualized over an entire field rather than only in one dimension for a single spot.

The structure of the thesis is as follows. In Section 2, we introduce the mathematical formulation of the inverse problem and discuss existing methods from the literature. In particular, we distinguish between the formulation with a fixed forward model and the formulation in which part of the kernel is also treated as unknown. In Section 3, we describe the least-squares algorithms and validate it on synthetic data. In Section 4, we present the results for real data, first in one dimension and thereafter in the two-dimensional extension. Finally, Section 5 contains the conclusions and a discussion of the main findings.

2 | Preliminaries & existing methods

Field layout, height profile and noise

As discussed in the introduction, our ultimate goal is to recover the true wafer height profile from level sensor measurements. The quantity that we are interested in finding is the unknown wafer height described by the unknown two-dimensional function

$$h(x, y) : \mathbb{R}^2 \to \mathbb{R}.$$

In this thesis, we will first and mainly focus on the one-dimensional setting and study the height profile only in the scan direction y. So, for a fixed value of x, we write the unknown height profile as a function of y. Physically, this corresponds to selecting one spot of the level sensor and considering the measurements obtained by that spot during a stroke in the y-direction. Henceforth, in the one-dimensional setting, for the unknown wafer height, we write:

$$h(y) : \mathbb{R} \to \mathbb{R}.$$

As we mentioned in the introduction, we will be considering a class of wafers for which we already know the field layout and topography, so that we can evaluate the accuracy of our methods and verify whether the methods are working properly. Now, in the left image in Figure 2.1 below, we can see the nominal layout of a single field in two dimensions. Note that the light blue areas are lower in height and we will refer to these areas as *trenches*. The dark blue areas are areas that are higher in height than the light blue areas, and we will refer to these as the *high areas*. In the right image in Figure 2.1 below, we can see the one-dimensional nominal field layout of a single field in the scan direction y, where we have fixed x such that the cross section intersects the higher dark blue areas as well.

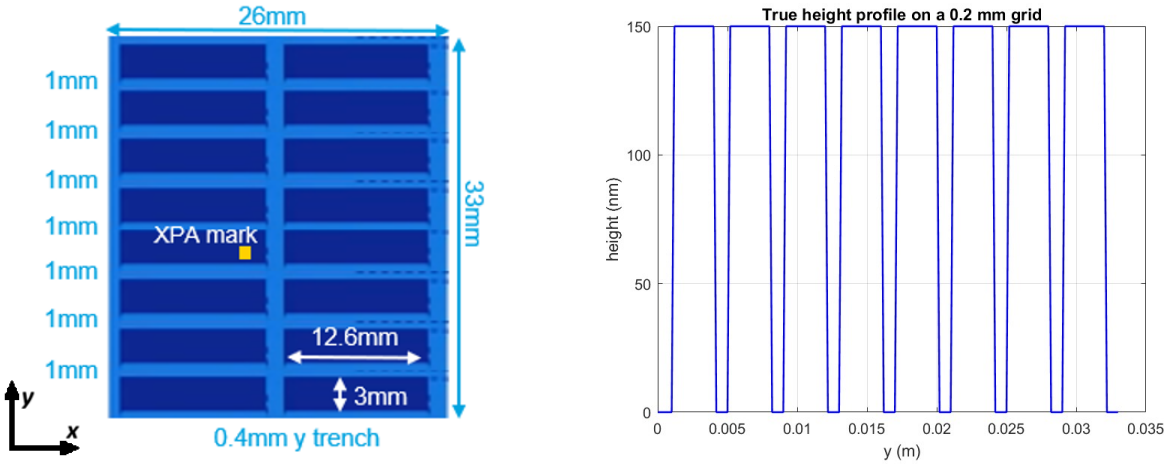


Figure 2.1: (Left) Nominal layout of a single field. (Right) Cross section of the nominal layout in the y-direction.

For the validation with synthetic data in Section 3.2, we want to simulate and use realistic noise in our model. For this reason, we have investigated which distribution the noise in the machine approximately follows. This is explained in more detail in Section 3.3 "Noise investigation" of the internship report [5]. We do this by looking at a very large sample of empirical noise values from an ASML machine. These noise values were obtained by measuring a bare wafer in the machine, which is a wafer without any topography on it. In particular, in this experiment, the level sensor measured the same exact spot on this wafer for 60 seconds without moving in the scan direction. If there was no noise present, the height measurement that the level sensor would output in this instance should naturally be a constant value for the whole time it is standing still. However, when we look at the empirical distribution of the height values reported by the level sensor for this experiment, we see an approximate normal distribution, see Figure 2.2. The sample standard deviation of this very large sample was equal to

$$4.72 \cdot 10^{-9} \text{ m} = 4.72 \text{ nm}.$$

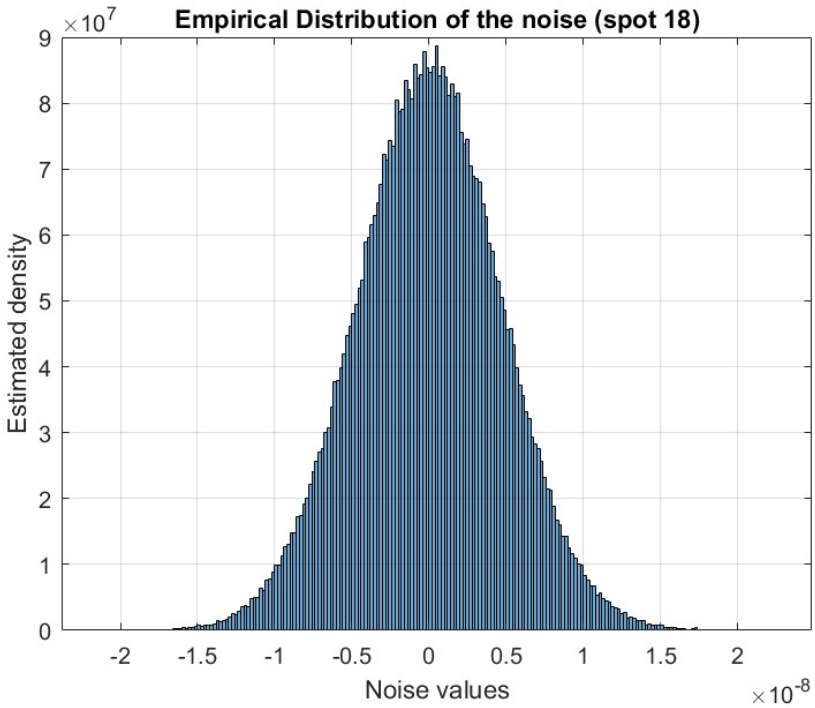


Figure 2.2: Empirical distribution of the noise in a machine.

With these preliminaries in place, this chapter is organized as follows:

- In Section 2.1, we formulate the inverse problem. There, we introduce the precise mathematical models that are used in this thesis and distinguish between the formulation with a fixed forward model and the formulation in which part of the kernel is also treated as unknown.
- In Section 2.2, we discuss existing methods from the literature and place the present work in that broader context. In particular, we will discuss why semi-blind deconvolution is a natural idea in the present setting, and why the structure of the wafer topography makes a least-squares approach especially attractive and natural here.

2.1 | Mathematical formulation of the inverse problem

Now, we will define a precise mathematical formulation of the inverse problems.

Parameterization of $h(y)$

As we have already discussed in the introduction, we do not try to recover an arbitrary one-dimensional function $h(y)$. Instead, we use the fact that the topography inside a selected field is known a priori to have a step profile with fixed step locations in the scan direction. This step profile can clearly be seen in the plot on the right in Figure 2.1. The domain of a field in the y-direction is $[0 \text{ mm}, 33 \text{ mm}) \subset \mathbb{R}$.

Now, let the step locations (in the y-dimension) be given by:

$$0 \text{ mm} = \theta_0 < \theta_1 < \ldots < \theta_{S-1} < \theta_S = 33 \text{ mm},$$

where S denotes the number of segments in the field. The number of segments S is equal to the number of trenches + the number of high areas. For the specific field layout that we consider, we have $S = 17$. The key assumption now is that the unknown height profile $h(y)$ is piecewise constant on these segments. This means that for each segment $j \in \{1, \ldots, S\}$ (with boundaries located at θ_{j-1}, θ_j), the corresponding height is given by a constant height value, denoted by η_j. Then we define $h(y)$ as:

$$h(y) = \sum_{j=1}^{S} \eta_j \, \mathbf{1}\{y \in [\theta_{j-1}, \theta_j)\}.$$

Clearly, since we assume the fixed step locations $\theta_0, \ldots, \theta_S$, the unknown height profile $h(y)$ is fully determined by the following vector:

$$\eta = [\eta_1, \ldots, \eta_S]^T \in \mathbb{R}^S.$$

Hence, the inverse problem is reduced to estimating the finite-dimensional parameter vector $\eta \in \mathbb{R}^S$.

The forward model

Here and throughout the remainder of this thesis, the symbol $*$ denotes the convolution operator. The first convolution that occurs when the level sensor measures $h(y)$ is:

$$(\mathcal{L} * h)(y),$$

where, analytically, we define the convolution by:

$$(\mathcal{L} * h)(y) = \int_{\mathbb{R}} \mathcal{L}(u)\, h(y-u)\, du.$$

Here, $\mathcal{L}$ models the level sensor optical response. So, for every position y, the value $(\mathcal{L}*h)(y)$ is a weighted average of nearby values of the original height profile h, with weights determined by the kernel $\mathcal{L}$. This is exactly why convolution with $\mathcal{L}$ smooths the original sharp step profile. The best model that we have of this kernel $\mathcal{L}$, describes it as:

$$\mathcal{L} = \mathcal{L}_{\text{ave}} + \mathcal{L}_{\text{CWE}}.$$

So the kernel $\mathcal{L}$ is made up of an averaging kernel $\mathcal{L}_{\text{ave}}$ and a CWE kernel $\mathcal{L}_{\text{CWE}}$, responsible for the curved wafer effect. For this level sensor response kernel $\mathcal{L}$, we have access to an ASML simulation tool that simulates the kernel in a discrete setting. We move to a discrete setting in Section 2.1.1. Let u denote the relative coordinate with respect to the current measurement location (the center of the spot). The simplest description that we can use for the averaging kernel is a uniform averaging kernel:

$$\mathcal{L}_{\text{ave}}(u) = \frac{1}{w}\mathbf{1}\left\{|u| \leq \frac{w}{2}\right\},$$

where $w > 0$ denotes the effective width of the level sensor footprint in the y-direction. This kernel simply averages the height profile over the footprint of the level sensor spot. A property that the kernel $\mathcal{L}_{\text{ave}}$ has to satisfy is that it integrates to one. The factor $\frac{1}{w}$ normalizes the kernel, so that indeed

$$\int_{\mathbb{R}} \mathcal{L}_{\text{ave}}(u)\, du = 1.$$

In the simulation tool $\mathcal{L}_{\text{ave}}$ can also have a more realistic trapezoidal shape (the shape used in the simulation tool depends on the specific type of level sensor selected in the tool), where the response is constant in the center of the footprint and decreases linearly near the edges. The important point for the mathematical formulation is that $\mathcal{L}_{\text{ave}}$ is a normalized footprint-intensity kernel that integrates to one.

The CWE part of the kernel, $\mathcal{L}_{\text{CWE}}$, can be described as the derivative of the footprint-weighted lever arm and is modeled using the product of the footprint intensity with the y-arm. In the idealized case with zero offset (this offset is denoted by the parameter τ), this y-arm is simply given by the relative coordinate u. Hence we define:

$$a(u) = u\, \mathcal{L}_{\text{ave}}(u),$$

where $a(u)$ corresponds to the 'y-arm times intensity' term. Then, in a continuous setting, the CWE kernel is obtained by applying a derivative in the scan direction:

$$\mathcal{L}_{\text{CWE}}(u) = \frac{d}{du}\left(u\, \mathcal{L}_{\text{ave}}(u)\right).$$

Equivalently, if D_y denotes a derivative kernel in the y-direction, this can be written as:

$$\mathcal{L}_{\text{CWE}}(u) = \left(D_y * a\right)(u).$$

Therefore, the total one-dimensional level sensor response kernel is written as:

$$\mathcal{L}(u) = \mathcal{L}_{\text{ave}}(u) + \mathcal{L}_{\text{CWE}}(u).$$

In the remainder of this thesis, these expressions should be interpreted as the analytical motivation for the structure of the kernel. In the actual numerical implementation, the corresponding discrete response is generated using the ASML simulation tool.

Note that for this continuous description above, the CWE kernel has zero integral:

$$\int_{\mathbb{R}} \mathcal{L}_{\text{CWE}}(u)\, du = 0.$$

This is also physically natural, since the CWE contribution should not create an offset for a perfectly flat height profile. Instead, it mainly affects the response near height transitions, which is exactly where the curved wafer effect becomes relevant.

Now, naturally, when the level sensor measures $h(y)$, we also measure some noise, which gives:

$$(\mathcal{L} * h) + \varepsilon,$$

where ε denotes the noise that occurs when the level sensor measures the wafer. As discussed in the preliminaries, we found from the noise investigation that the noise was approximately normally distributed with a sample standard deviation of 4.72 nm.

Finally, $\mathcal{F}$ models the driver filter that is applied to the signal in the machine. This is a signal processing step that takes place after the level sensor response. In our specific case, it is an FFT filter with the purpose of suppressing high-frequency components in the level sensor signal. In particular, such filtering changes the amplitude of spatial features and can also affect the behavior near sharp transitions.

With this, we have that the measured signal z is modeled as follows:

$$z = \mathcal{F}\left((\mathcal{L} * h) + \varepsilon\right). \tag{2.1}$$

So the forward model first smooths the original sharp profile through $\mathcal{L}$ and then filters the resulting signal through $\mathcal{F}$.

2.1.1 | Indicator matrix and discrete height profile

We now discretize the field in the scan direction, in accordance with the internal 0.2 mm grid for the simulation tool, and generate a grid:

$$y_1, \ldots, y_n,$$

with:

$$y_1 = 0 \text{ mm}, \qquad y_n = 33 \text{ mm}, \qquad y_i - y_{i-1} = 0.2 \text{ mm}, \quad \text{for all } i = 2, \ldots, n.$$

This yields $n = 166$ grid points. Since we are in a discrete setting now, a discretized height profile is no longer a continuous function $h(y)$, but any discretized piecewise constant height profile $\boldsymbol{h} \in \mathbb{R}^n$ is now a vector of height values at the grid points:

$$\boldsymbol{h} = [h(y_1), \ldots, h(y_n)]^T \in \mathbb{R}^n.$$

Using the known segment boundaries/step locations $\theta_0, \ldots, \theta_S$, we define the indicator matrix:

$$I_{\text{seg}} \in \mathbb{R}^{n \times S}$$

as:

$$[I_{\text{seg}}]_{i,j} = \begin{cases} 1, & \text{if } y_i \text{ lies in segment } j, \\ 0, & \text{otherwise.} \end{cases}$$

So the j-th column of I_{seg} is simply the indicator vector of segment j for the grid. The role of I_{seg} is fundamental. It allows us to write any discretized piecewise constant height profile $\boldsymbol{h} \in \mathbb{R}^n$ with the prescribed segment boundaries on our grid in the following form:

$$\boldsymbol{h} = I_{\text{seg}}\, \boldsymbol{\eta}, \qquad \boldsymbol{\eta} = [\eta_1, \ldots, \eta_S]^T \in \mathbb{R}^S. \tag{2.2}$$

This means that once the segment boundaries are fixed, the entire unknown profile on our grid is determined by the vector $\boldsymbol{\eta}$ of segment heights. In this thesis, I_{seg} is the matrix that translates the unknown parameter vector $\boldsymbol{\eta} \in \mathbb{R}^S$ into the corresponding sharp step profile on the internal 0.2 mm grid.

The nominal profile/true height profile

For the wafers we considered in this thesis, the nominal height profile is an alternating two-level profile. Hence, the nominal segment-height vector is of the form:

$$\boldsymbol{\eta}_{\text{true}} = [\eta_1, \eta_2, \eta_1, \ldots, \eta_2, \eta_1]^T \in \mathbb{R}^S,$$

where, in the experiments considered here, unless explicitly stated otherwise, we have:

$$\eta_1 = 0 \text{ nm}, \qquad \eta_2 = 150 \text{ nm}.$$

Using (2.2), this yields the nominal profile/true height profile $\boldsymbol{h}_{\text{true}} \in \mathbb{R}^n$:

$$\boldsymbol{h}_{\text{true}} = I_{\text{seg}}\, \boldsymbol{\eta}_{\text{true}}.$$

For the synthetic-data experiments, this vector is what we consider to be the actual ground truth. For the real-data experiments however, it serves as the nominal reference pattern with which the estimated profile is compared. This is because there may be a global shift in the real-data, due to which we cannot directly compare the nominal reference profile with the height profile that we estimate.

Discrete forward operator

We defined the operators $\mathcal{L}$ and $\mathcal{F}$ in a continuous setting and obtained the forward model as described in (2.1). However, since we are now in a discrete setting, we introduce the discrete operators $\mathcal{L}_n$ as the discrete version of $\mathcal{L}$ and $\mathcal{F}_n$ as the discrete version of $\mathcal{F}$. We will also use a discrete function denoted by $\mathcal{S}_n$.

The continuous level sensor response operator $\mathcal{L}$ is now represented by a discrete operator

$$\mathcal{L}_n : \mathbb{R}^n \to \mathbb{R}^n.$$

$\mathcal{L}_n$ maps a sharp height profile on the internal 0.2 mm grid to the corresponding level sensor response on the same grid. In line with the continuous description, this discrete response consists of an averaging part and a CWE part:

$$\mathcal{L}_n(\boldsymbol{h}) = \mathcal{L}_{\text{ave},n}(\boldsymbol{h}) + \mathcal{L}_{\text{CWE},n}(\boldsymbol{h}).$$

In the simulation tool, the averaging part is obtained by a discrete convolution with the normalized footprint-intensity kernel. As mentioned, the shape of this discrete kernel depends on the type of level sensor. The simulation tool in MATLAB uses the function `conv` for this discrete convolution.

For the CWE part of $\mathcal{L}_n$, we have three different models available in the simulation tool. In the first model, named the linear CWE response model, the CWE part is obtained by first computing a discrete derivative (using `gradient` in MATLAB) of the height profile in the y-direction and then convolving this derivative with the discrete CWE kernel. The discrete CWE kernel is the y-arm times the normalized footprint-intensity kernel. This first model can be thought of as a discrete version of the continuous CWE model we described for $\mathcal{L}_{\text{CWE}}$. The term 'linear' in the name of this first model refers to the fact that the 'ears' (the CWE artefact as seen in the top right plot in Figure 3.1) scale linearly with step height.

Additionally, there is a more realistic empirical model including optical interference, and an alternative model which reportedly has a worse match with real data than version 2. We tested these three different models on synthetic data, and for our specific profile, the linear CWE response yielded more accurate results than the other models. Accordingly, this is the one we proceed with.

The continuous driver filter operator $\mathcal{F}$ is represented in the discrete setting by

$$\mathcal{F}_n : \mathbb{R}^n \to \mathbb{R}^n.$$

For our specific setup, the simulation tool for $\mathcal{F}_n$ performs a discrete FFT implementation on the signal.

Finally, for real data we also need to map the simulated signal from the internal 0.2 mm grid to the actual measured within-field coordinates. We denote this discrete sampling/interpolation operator by

$$\mathcal{S}_n : \mathbb{R}^n \to \mathbb{R}^m,$$

where m is the number of measured data points inside the selected field. For synthetic data, the measurements are generated directly on the internal 0.2 mm grid, so $m = n$ and

$$\mathcal{S}_n = I_n.$$

For real data, however, the measured data is only available on a coarser grid, with approximately 65 or 66 points inside a field. In that case, $\mathcal{S}_n$ is the interpolation step from the internal simulation grid to the measured within-field coordinates.

With these discrete operators, the discrete forward model for the measurements becomes:

$$z_{\text{meas}} = \mathcal{S}_n \left(\mathcal{F}_n \left(\mathcal{L}_n \left(\boldsymbol{h} \right) + \varepsilon_n \right) \right), \qquad \varepsilon_n \sim \mathcal{N}_n(0, \sigma^2 I). \tag{2.3}$$

Substituting (2.2) into this discrete forward model yields:

$$z_{\text{meas}} = \mathcal{S}_n \left(\mathcal{F}_n \left(\mathcal{L}_n \left(I_{\text{seg}} \, \boldsymbol{\eta} \right) + \varepsilon_n \right) \right).$$

Since the discrete forward model is linear in the height parameters $\boldsymbol{\eta}$, we can define the matrix:

$$\Phi = \mathcal{S}_n \left(\mathcal{F}_n \left(\mathcal{L}_n \left(I_{\text{seg}} \right) \right) \right),$$

such that the final discrete forward model for the measurements can be written compactly as:

$$z_{\text{meas}} = \Phi \boldsymbol{\eta} + \varepsilon_n', \qquad \text{where } \varepsilon_n' = \mathcal{S}_n \left(\mathcal{F}_n \left(\varepsilon_n \right) \right).$$

This matrix Φ serves as the design matrix of the least-squares problem. Its j-th column is obtained by taking the j-th column of I_{seg}, applying the level sensor response $\mathcal{L}_n$, then the filter $\mathcal{F}_n$, and finally the sampling/interpolation operator $\mathcal{S}_n$. So intuitively, each column j of Φ describes how segment j contributes to the measured signal after passing through the entire forward model.

Problem P1: fixed forward model

We first consider the case in which the forward model is assumed fully known and fixed. In this case, the only unknowns are the segment heights collected in the vector $\boldsymbol{\eta}$. Then, the inverse problem becomes the following least-squares problem:

$$\textbf{(P1)} \qquad \min_{\boldsymbol{\eta} \in \mathbb{R}^S} \left\| z_{\text{meas}} - \Phi \boldsymbol{\eta} \right\|^2, \tag{2.4}$$

where $\|\cdot\|$ denotes the Euclidean norm. This problem has minimizer:

$$\hat{\boldsymbol{\eta}} = \arg\min_{\boldsymbol{\eta} \in \mathbb{R}^S} \left\| z_{\text{meas}} - \Phi \boldsymbol{\eta} \right\|^2. \tag{2.5}$$

This is the inverse problem considered throughout most of this thesis. Once the minimizer $\hat{\boldsymbol{\eta}}$ has been found, the sharp reconstructed/estimated height profile is given by:

$$\boldsymbol{h}_{\text{est}} = I_{\text{seg}} \, \hat{\boldsymbol{\eta}}.$$

Problem P2: partially unknown forward model

The second inverse problem considered in this thesis is the case in which part of the kernel is also treated as unknown. More specifically, we allow the level sensor response $\mathcal{L}_n$ to depend on a parameter τ, which is related to a shift in y of the kernel $\mathcal{L}_n$. So we now write:

$$\mathcal{L}_{n,\tau}.$$

Therefore, the measurement model becomes:

$$z_{\text{meas}} = \mathcal{S}_n\left(\mathcal{F}_n\left(\mathcal{L}_{n,\tau}\left(I_{\text{seg}}\,\eta\right) + \varepsilon_n\right)\right).$$

Correspondingly, the design matrix Φ then also becomes dependent on τ:

$$\Phi(\tau) = \mathcal{S}_n\left(\mathcal{F}_n\left(\mathcal{L}_{n,\tau}\left(I_{\text{seg}}\right)\right)\right).$$

Now both the height vector η and the kernel parameter τ are unknown. This yields the following semi-blind inverse problem:

$$\textbf{(P2)} \qquad \min_{\eta\in\mathbb{R}^S,\,\tau\in\mathbb{R}} \left\|z_{\text{meas}} - \Phi(\tau)\eta\right\|^2, \tag{2.6}$$

with minimizer:

$$(\hat{\eta},\hat{\tau}) = \operatorname*{arg\,min}_{\eta\in\mathbb{R}^S,\,\tau\in\mathbb{R}} \left\|z_{\text{meas}} - \Phi(\tau)\eta\right\|^2. \tag{2.7}$$

So, in Problem **P2**, we simultaneously estimate the sharp wafer height profile and part of the forward model through the parameter τ. This is why **P2** is referred to as a semi-blind deconvolution problem.

The distinction between **P1** and **P2** will be central in the remainder of the thesis. Problem **P1** leads to a linear least-squares algorithm, whereas Problem **P2** leads to a nonlinear optimization problem in which the unknown height parameters still enter linearly for fixed τ. This structure will be exploited later on in Section 3 when discussing the reconstruction algorithms.

Next, we will place **P1** and **P2** in a broader context, namely the literature of currently existing blind and semi-blind deconvolution methods.

2.2 | Literature review

Now that we have formulated the inverse problems **P1** and **P2**, we turn to the literature on deconvolution and blind deconvolution. Naturally, there is a very large amount of literature on inverse problems of this type. Most of this literature comes from image and signal processing, where the goal is often to recover a sharp image from a blurred photograph. Of course, our setting is quite different from this. We are not dealing with natural images but with highly structured wafer topography that is largely step-like and piecewise constant, whereas photographs are likely to have more smooth transitions. Nevertheless, many of the central ideas and overarching structures and relations from that literature are still very relevant here.

A useful first distinction is that between *non-blind* and *blind* deconvolution. In non-blind deconvolution, the forward operator or blur kernel is assumed known and fixed, and only the latent sharp signal is estimated. In blind deconvolution, the blur kernel itself is also unknown and is estimated from the data together with the latent signal. In the terminology of this thesis, Problem **P1** is a non-blind deconvolution problem, whereas Problem **P2** is a semi-blind deconvolution problem, since only a low-dimensional parameter (namely in $\mathbb{R}$) of the kernel is treated as unknown.

Before discussing the individual papers, it is useful to keep one general point in mind. In the standard image-deblurring literature, the latent object is usually a general natural image, and therefore strong priors are needed in order to regularize the problem. In our setting, however, we already assume that the field layout is known and that the latent true profile inside a field is piecewise constant with known step locations. This means that a substantial part of the regularization that image-deblurring methods must impose explicitly is, in our case, already built into the model class itself. This difference will return throughout the discussion below. A compact overview of the main papers, their image priors, kernel priors and estimation strategies is given in Table 2.1, located at the end of this section.

Levin et al. (2009): why naive joint MAP estimation (MAP$_{x,k}$ estimation) often fails

A very important and foundational paper for understanding blind deconvolution is the work by Levin et al. [9]. The central question in this paper is the following. If we have a blurred observation:

$$y = k * x + n,$$

where x is the unknown sharp image, k is an unknown blur kernel and n is i.i.d. Gaussian noise, why can we not simply estimate both the latent sharp image x and the blur kernel k by straightforward joint MAP$_{x,k}$ estimation? This means that we would estimate x and k together, at the same time.

At first sight, this seems like a straightforward and natural approach. This approach entails finding x, k that maximizes:

$$p(x, k|y) \propto p(y|x, k)p(x)p(k), \tag{2.8}$$

where the p's are the respective probability density functions. Intuitively, this makes sense: we seek to find the most likely original image *and* blur kernel given the observation y. In other words, we have:

$$(\hat{x}, \hat{k}) = \arg\max_{x,k} p(x, k|y). \tag{2.9}$$

However, [9] shows why there are limitations to this exact method. A uniform prior on the kernel k is assumed for the sake of simplicity of exposition. Moreover, the log-likelihood function (in the literature often referred to as 'data fitting term' or 'data term') describes the distribution of observations given the true image and kernel. In the case of i.i.d. Gaussian noise with variance σ^2, we have:

$$\log p(y|x, k) = -\lambda||k * x - y||^2 + C_1$$
$$p(y|x, k) = e^{-\lambda||k*x-y||^2} e^{C_1},$$

where:

$$\lambda = -\frac{1}{2\sigma^2}.$$

This expression is directly related to least squares. Maximizing the Gaussian likelihood is equivalent to minimizing the squared residual

$$\|k * x - y\|^2.$$

This is the same data fitting structure as in our semi-blind least-squares formulation in (2.7). In the notation of this thesis, the observed blurry signal y corresponds to z_{meas}, the latent sharp image x corresponds to the parameterized height profile $I_{\text{seg}}\eta$, and the blurred prediction $k * x$ corresponds to $\Phi(\tau)\eta$. Hence, the least-squares term in (2.7),

$$\|z_{\text{meas}} - \Phi(\tau)\eta\|^2,$$

can be viewed as the Gaussian noise data term for the present wafer height inverse problem. The difference with the MAP formulation in the blind-deconvolution literature is that MAP estimation also includes priors on the latent image and possibly on the kernel. After taking the negative logarithm of the posterior, these priors become additional regularization terms. In our formulation, we do not add such explicit image gradient priors. Instead, a large part of the prior structure is already built into the model by assuming a piecewise constant height profile with known step locations and, in Problem **P2**, by allowing the kernel to vary only through the scalar parameter $\tau \in \mathbb{R}$.

In [9], for the prior distribution $p(x)$ of the original sharp image, the following choice is made:

$$\log p(x) = -\sum_i |g_{x,i}(x)|^\alpha + |g_{y,i}(x)|^\alpha + C_2 \tag{2.10}$$

$$p(x) = e^{-\sum_i |g_{x,i}(x)|^\alpha + |g_{y,i}(x)|^\alpha} e^{C_2}. \tag{2.11}$$

Where $g_{x,i}(x)$ is a finite difference in the horizontal direction of pixel i and $g_{y,i}(x)$ is a finite difference in the vertical direction of pixel i. C_1 and C_2 are normalization constants.

Now, observe that larger absolute values of $g_{x,i}$ and $g_{y,i}$ will make $p(x)$ smaller, and smaller absolute values of $g_{x,i}$ and $g_{y,i}$ will make $p(x)$ larger. Since $p(x)$ is maximized in (2.8), this means that this choice of prior for $p(x)$ penalizes images with a lot of high gradients, and favors images that have mostly sparse gradients and as a result a smaller sum of finite differences. The reason for this choice in [9] is that natural images tend to have mostly zero gradients. For the height profile of a wafer, this is also true: we have mostly flat regions (where the gradient is zero) and only a few steps/jumps where the gradient is high. So this choice for the prior $p(x)$ could at least potentially be meaningful in our approach.

Given these assumptions on the likelihood function and the priors on k and $p(x)$, we can substitute the expressions above and formulate our objective as:

$$\begin{aligned}\hat{x}, \hat{k} &= \arg\max_{x,k} p(x,k|y) = \arg\max_{x,k} p(y|x,k)p(x)p(k) \\ &= \arg\max_{x,k} e^{-\lambda||k*x-y||^2} e^{C_1} e^{-\sum_i |g_{x,i}(x)|^\alpha + |g_{y,i}(x)|^\alpha} e^{C_2}.\end{aligned}$$

Since e^{C_1} and e^{C_2} are simply constants, this is equivalent to:

$$\begin{aligned}\hat{x}, \hat{k} &= \arg\max_{h,k} e^{-\lambda||k*x-y||^2} e^{-\sum_i |g_{x,i}(x)|^\alpha + |g_{y,i}(x)|^\alpha} \\ &= \arg\min_{x,k} \lambda||k*x-y||^2 + \sum_i |g_{x,i}(x)|^\alpha + |g_{y,i}(x)|^\alpha,\end{aligned}$$

where the last step naturally follows from the fact that we want to minimize what is multiplied by -1 in both of the exponents and take the logarithm. In [9], it is shown in claim 1 that if we do not constrain the kernel and the intensities of the blurred and sharp images, $\text{MAP}_{x,k}$ estimation will output an image with no gradients (i.e. a flat image) and a kernel with infinite norm. This gets better when these constraints are in fact applied, a few examples will now follow where this is the case.

A few one-dimensional examples are discussed in which we look at the performance of $\text{MAP}_{x,k}$ estimation, in which the value of α also plays a role. In the top row in Figure 2.3, examples of original 1D signals and

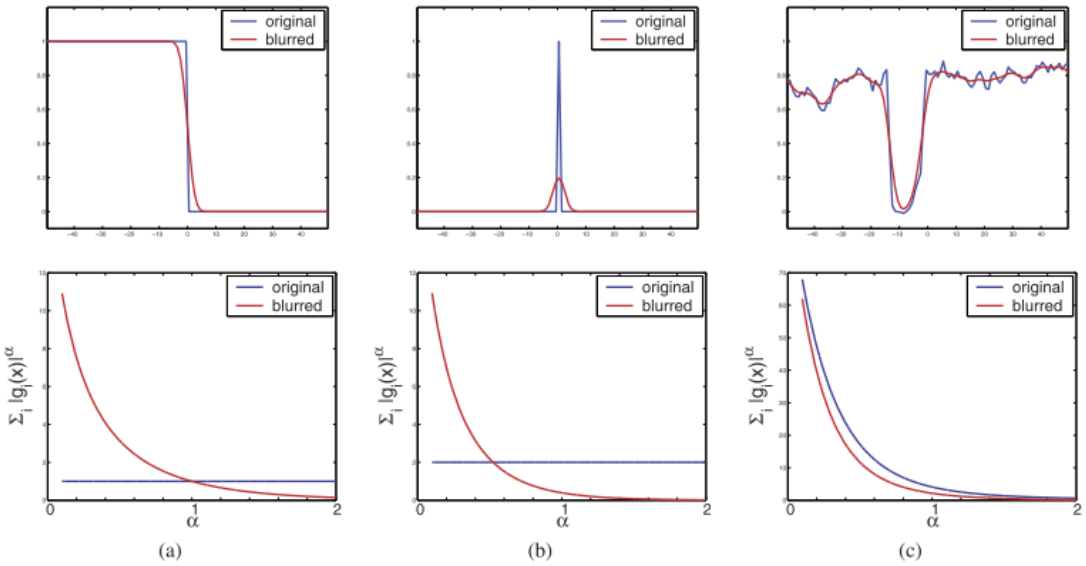


Figure 2.3: Top row: 1D sharp and blurred signals. Bottom row: sum of gradients as a function of α. Figure from [9].

the corresponding signals that have been convolved by a Gaussian kernel k' are shown. On the y-axes in the plots on the bottom row, we have $\sum_i |g_i(x)|^\alpha$ for different values of α. Note that since we are working in one dimension, we can only look at finite differences in one dimension. Now, we compare two cases, namely the true kernel case:

$$y = k' * x,$$

versus the delta kernel (no blur) case:

$$y = \delta * y.$$

On an intuitive level, the true kernel case, tells us that the original image is x and this image got convolved by k', whereas the delta kernel case tells us that the original image is actually equal to the observed blurred image y. Note that there is no noise involved in this example, because of this, we have for both cases that:

$$\lambda||k * x - y||^2 = \lambda||k' * x - k' * x||^2 = 0,$$
$$\lambda||k * x - y||^2 = \lambda||\delta * x - \delta * y||^2 = 0.$$

Our objective in this one-dimensional setting is:

$$\arg\min_x \lambda||k * x - y||^2 + \sum_i |g_i(x)|^\alpha.$$

Since we just showed that the likelihood is zero, we find this to be equivalent to:

$$\arg\min_x \sum_i |g_i(x)|^\alpha. \tag{2.12}$$

Therefore, in these cases, $\text{MAP}_{x,k}$ estimation is simply equivalent to minimizing the sum of the gradients from the prior, so we choose the original image based on which of the two, the blurred or the original, has a sparser distribution of gradients.

Then, turning to Figure 2.3, we can see in the leftmost example of a sharp step profile, that for $\alpha < 1$, we indeed pick the original image and the true kernel case, over the no blur explanation, but for the other original image profiles, this becomes a bigger challenge for nearly all values of α.
This point is directly relevant for this thesis as well. Our original signal is clearly not a natural image, but it does have a strong step structure. This means that, locally around true height steps, this sparse-gradient result is more favorable for us than it is for generic photographic images. However, the central message still remains relevant: once both the latent signal and the kernel are allowed to vary freely, the inverse problem becomes highly ill-posed, and one should be very cautious with this.

So in conclusion, the straightforward $\text{MAP}_{x,k}$ optimization for most natural images and with typical (sparse) image priors often fails and the joint $\text{MAP}_{x,k}$ estimate converges to the no blur case. Levin et al. also argue that the failure of $\text{MAP}_{x,k}$ is not merely a matter of having chosen the wrong sparse prior. Their analysis shows that the problem is more structural, the prior alone does not fix the ill-posedness of the joint estimation problem. Without strong constraints, there are too many x, k pairs that can explain the blurry observation y. Moreover, they argue that the choice of estimator is more important than the exact choice of prior. This finding motivates the next section in which the estimation strategy changes in order to try to tackle this issue with $\text{MAP}_{x,k}$ estimation.

2.2.1 | MAP_k estimation

One way to get around this issue of the no-blur case being the prevalent solution, is to change the estimation method. Therefore, in [9] and [8], MAP_k estimation is introduced. This approach focuses on kernel k estimation first and makes use of the fact that the kernel often has limited support relative to the image size/number of measurements; the number of unknowns in the image grows with image size, whereas the kernel support is typically relatively small. In MAP_k estimation, we first use a MAP_k estimator to recover the kernel and thereafter solve for the latent image x using a non-blind deconvolution algorithm. A MAP_k estimator selects:

$$\hat{k} = \arg\max_k p(k|y),$$

where:

$$p(k|y) = p(y|k)p(k)/p(y).$$

We obtain $p(y|k)$ by marginalizing over x:

$$p(y|k) = \int p(x, y|k)dx.$$

In [8], it is shown that MAP_k can succeed where $\text{MAP}_{x,k}$ fails. Moreover, as already mentioned, the choice of estimator (so MAP_k vs $\text{MAP}_{x,k}$) is more important than the form of the image prior. This fact underpins several methods in which there are alternating steps or two-stage optimization methods, ensuring the kernel is well-constrained before final image recovery.

Levin et al. (2011): efficient marginal likelihood optimization

The follow-up paper by Levin et al. [8] takes the next logical step. If MAP_k estimation is conceptually the right thing to do, then how can one actually optimize it in practice? The difficulty is of course that the MAP_k estimator contains an integral over all possible latent sharp images x, and that integral is generally speaking intractable.

The contribution of [8] is that it derives a practical approximation to this MAP_k idea and shows that it can be optimized with only a modest modification of standard alternating blind-deconvolution algorithms. The paper formulates the problem in an expectation-maximization (EM) framework. This EM framework is an iterative optimization framework for finding MAP estimates for problems in which the model depends on unobserved latent variables. In this paper, the latent image is treated as a hidden variable and the algorithm alternates between estimating information about this hidden image (in an E-step) and updating the kernel based on that information (an M-step). In more detail, at a high level one alternates between:

- an E-step, in which one estimates a distribution over the latent image for the current kernel, summarized by a mean image and a covariance,
- and an M-step, in which one updates the kernel and finds the best kernel using not only the mean latent image estimate, but also the covariance around it.

This is an important difference compared to standard $\text{MAP}_{x,k}$ alternating minimization. The fact that here the covariance around the mean latent image is taken into account is what distinguishes the approach from the $\text{MAP}_{x,k}$ style approach.

As for the relevancy to this thesis, our semi-blind problem is not exactly a MAP_k method, and we do not compute a covariance over all possible sharp profiles. However, in an abstract way, the paper is conceptually related: we do exploit structure in the problem in order to avoid a completely free or unstable joint estimation. In our case, the structure is even stronger, because the sharp profile is parameterized by only a finite-dimensional vector of segment heights and the kernel uncertainty is restricted to a single scalar parameter τ, so this is in line with the idea that the kernel should be well constrained.

Fergus et al. (2006): heavy-tailed image gradient priors

The paper by Fergus et al. [2] is one of the classical papers that made blind deconvolution with natural-image priors practically successful. Their setting is single-image camera-shake removal. A central observation is that camera-shake kernels are often quite complex and sharp, and therefore cannot be modeled well by overly simple parametric kernels or by purely frequency-domain assumptions.

The prior on the latent image in [2] is based on heavy-tailed natural-image gradient statistics. More specifically, the gradient distribution is approximated by a mixture of Gaussians. This is a convenient choice, because heavy-tailed distributions are empirically realistic for natural images and the mixture representation makes variational inference tractable. For the kernel, the paper uses a sparsity promoting prior with non-negativity constraints. In the formulation this becomes a mixture of exponential distributions on the kernel entries. All entries are required to be positive and the sparsity prior promotes zero values.

The estimation method is variational Bayesian. Roughly speaking, this means that instead of optimizing only one single sharp image and one single kernel directly, the method approximates a posterior probability distribution over the unknowns and updates this approximation iteratively and extracts the kernel from this approximate posterior distribution. This is one of the reasons why the method

succeeds where naive joint MAP attempts can fail. In fact, the paper explicitly reports that straightforward MAP-style optimization produced poor behavior such as a two-tone image or the delta-kernel solution.

Another important strategy of Fergus et al. is the coarse-to-fine strategy. Estimation is first performed at coarse scales and then refined progressively at finer scales. This helps avoid local minima and makes it possible to recover larger blur kernels than would be feasible at full resolution from the beginning. Additionally, the method uses a user-selected patch rich in edge structure and asks the user for an approximate kernel size and orientation. So the algorithm is not fully automatic in a strict sense.

For this thesis, the paper from Fergus et al. is relevant mainly as a foundational blind-deconvolution paper that shows how strong latent image priors, a Bayesian approach and coarse-to-fine optimization can together yield a practical success. At the same time, there are clear reasons why we should not directly adopt this type of method. Their prior is specifically tailored to natural image gradients, their setting is full image blind deblurring, and the method requires much more estimation machinery than is necessary in our more structured wafer height profile setting. Still, the central message remains relevant: once the kernel is not fully known, uncertainty in the latent image must be handled carefully, and it is often not enough to optimize a single sharp image and a single kernel jointly.

Shan, Jia and Agarwala (2008)

The paper by Shan, Jia and Agarwala [12] is interesting because it starts with an analysis of what causes ringing artifacts in deconvolution, which are ripples that can be either light or dark and occur near strong edges after deconvolution. The common belief is that ringing is mainly a so-called Gibbs phenomenon which is caused by finite Fourier representations near step edges being unable to model these steps well enough. Shan et al. argue that this is not the main explanation since they found finite Fourier representations able to model steps in natural images. Instead, they show that ringing is caused primarily by two things: image noise and kernel-estimation errors.

Based on this analysis, they propose a unified probabilistic model for both blind and non-blind deconvolution. The model explicitly encodes the idea that noise should be spatially random rather than structured. For the prior on the latent image, Shan et al. use both a global prior and a local prior. The global prior is based on the heavy-tailed distribution of gradients from natural images, but instead of using the same mixture-of-Gaussians prior as Fergus et al. [2], they fit a piecewise continuous function to the curve of logarithmic density of image gradients. The reason they give is practical: the mixture-of-Gaussians representation becomes difficult to optimize once the logarithm of the prior enters the MAP objective. So this is already an example of a paper in which not only the prior itself matters, but also whether the chosen representation can be optimized stably and efficiently.

The local prior uses the blurred image itself to detect locally smooth regions. In such regions, the recovered latent image should also remain smooth, and this helps suppress ringing artifacts that would otherwise appear. The paper shows that this local prior can suppress ringing not only in the smooth region itself, but also more globally due to the footprint of the blur kernel.

The resulting optimization alternates between kernel refinement and latent image restoration until it converges. So again we can see the structure of alternating blind-deconvolution methods, but this time with a more carefully engineered probabilistic model made specifically to suppress ringing artifacts.

For this thesis, the paper from Shan et al. is also particularly relevant because our wafer height profile also contains sharp step transitions. So although a discussion of ringing, structured residuals and artifact suppression is more directly relevant here than some of the more purely photograph based papers, their model is still designed for general natural images and unknown motion blur and not for very structured piecewise constant wafer profiles. So while the insight about ringing and model mismatch is useful, the full machinery of the paper would probably be unnecessarily heavy and overly complicated for our present more structured problem.

Cho and Lee (2009)

The paper by Cho and Lee [1] takes a different route, since their main goal is speed. They observe that many blind-deconvolution methods alternate between latent-image estimation and kernel estima-

tion, but that both of these steps are often computationally heavy. Their contribution is to accelerate both.

The key idea is to introduce a prediction step. Instead of relying on an expensive latent-image prior in every iteration, they predict strong salient edges from the current latent image estimate and use these predicted edges primarily for kernel estimation. In other words, the method tries to explicitly isolate the part of the image that is most informative for identifying the blur kernel. This is motivated by the observation that kernel estimation is mainly driven by strong edges anyway, whereas small details and artefacts are more likely to confuse the estimation.

After edge prediction, they use a deconvolution model with a Gaussian-type regularization for the latent image. The point is that once the strong edges are predicted and fed into kernel estimation, one no longer needs an extremely sophisticated latent image prior at that stage. This makes the overall method much faster.

A second important idea in the paper is to formulate kernel estimation in the space of image derivatives rather than directly with pixel values. The paper shows that this leads to a smaller condition number of the numerical system and also reduces the number of FFTs needed per conjugate gradient iteration. The paper reports very substantial speed gains compared to earlier methods while retaining comparable deblurring quality.

For our specific setting, the paper by Cho and Lee is interesting because our signal is also dominated by a limited number of strong step transitions. So the idea that kernel estimation should focus primarily on strong edges is clearly relevant. At the same time, the method remains very image-processing oriented, whereas in this thesis we prefer to exploit the known field structure directly.

Krishnan and Fergus (2009)

The paper by Krishnan and Fergus [6] is a paper on non-blind deconvolution with hyper-Laplacian priors. This makes it especially relevant to Problem **P1**, where the forward model is treated as fixed.

Their starting point is the empirical fact that natural image gradients are even more heavy-tailed than a Laplacian distribution. A hyper-Laplacian prior of the form:

$$p(x) \propto e^{-k|x|^{\alpha}}, \qquad 0.5 \leq \alpha \leq 0.8,$$

fits image-gradient statistics significantly better than a Gaussian or standard Laplacian model. However, the difficulty is that this makes the resulting optimization usually quite slow.

The main contribution of the paper is algorithmic: they develop a very fast method for non-blind deconvolution under such hyper-Laplacian priors. Their method is based on half-quadratic splitting and alternating minimization. The non-convex part becomes separable over pixels, and then two solution strategies are proposed: a lookup-table approach and analytic solutions for the special cases $\alpha = 1/2$ and $\alpha = 2/3$. In this way, they obtain deconvolution quality comparable to much slower iteratively reweighted least-squares based approaches, but at a fraction of the computational cost.

This paper is relevant conceptually for our thesis, even though we do not adopt its precise method. It shows two things. Firstly, the choice of prior still matters in non-blind deconvolution, because heavier-tailed priors preserve edges better than Gaussian or Laplacian ones. Edges have very high gradients due to the sudden jumps so a distribution with heavy tails assigns higher probabilities to these higher gradients than distributions with less heavy tails. Secondly, computational tractability of a prior is also very important, not just statistical realism. A prior can be very accurate statistically, but if the resulting optimization is very slow, then it has limited practical value.

For our present work there is an important difference. In our formulation, the latent signal is not a general image but a piecewise constant profile with known segment boundaries. Because of this, much of the edge-preserving behavior that Krishnan and Fergus need to enforce through a hyper-Laplacian prior is, in our setting, already encoded directly in the parameterization. This is one of the reasons why a plain least-squares method can already work so well for us.

Nah et al. (2017) and Kupyn et al. (2018): deep learning and kernel-free methods

More recent work in deblurring moves away from explicit priors and explicit kernel estimation and instead learns the mapping from blurry to sharp images directly from data. Two important examples in this direction are the papers by Nah et al. [10] and Kupyn et al. [7].

Nah et al. propose a deep multi-scale convolutional neural network for dynamic scene deblurring. Their motivation is that many classical methods depend on somewhat simple blur kernel parameterizations, such as locally linear or partially uniform blur, and that these assumptions fail in dynamic scenes containing more sophisticated things like multiple moving objects, camera motion, occlusions and depth variation. Their network works in a kernel-free, end-to-end way: it takes a blurry image as the input and directly predicts a sharp image, without explicitly estimating a blur kernel. The multi-scale architecture is designed to mimic coarse-to-fine optimization, and the training uses a large dataset of realistic blurry and sharp pairs obtained by averaging high-speed camera frames.

This is an important development because it bypasses many of the classical difficulties of kernel estimation. In particular, Nah et al. explicitly say that the method avoids artifacts typically caused by inaccurate kernel estimates.

DeblurGAN by Kupyn et al. goes one step further in the direction of image-to-image translation. It formulates deblurring as a so-called conditional generative adversarial network (GAN) problem and combines adversarial loss with perceptual loss to be the total loss. Instead of using classical loss functions like mean squared error or mean absolute error alone, the model is trained so that the restored image is both perceptually close to a sharp image and difficult for the discriminator (which is a network) to distinguish from real sharp images. They also propose a synthetic motion blur generation scheme based on random trajectories in order to create realistic datasets.

These deep learning based methods are important to mention because they represent a fundamentally different approach and are recently also becoming more prominent and more widely used. The prior is no longer written down explicitly in analytic form, but instead, it is learned implicitly from very large datasets. This can give very strong empirical results. For this thesis however, there are several reasons why this is not very feasible. Firstly, these methods require large amounts of training data containing matched pairs, which is usually not available in our setting. Secondly, these methods are specifically designed for general natural image blur, not for the step profiles we expect to encounter in wafer topography. Thirdly, these methods are also much less interpretable and explainable than a least-squares or a semi-blind inverse-problem formulation based on a physical model, which is very important in practice. Since one of the main goals of this thesis is also to understand the inverse problem mathematically and to exploit known physical and geometrical structure, it is safe to say a model-based approach is more appropriate.

Position of the present work

The literature discussed above suggests the following overall picture. If the forward model is assumed fixed and is sufficiently reliable, then non-blind deconvolution by least squares (or regularized least squares) is a natural starting point. If part of the kernel is unknown, then semi-blind methods become relevant, but completely free joint estimation of latent signal and kernel is often unstable or biased toward trivial explanations. In that case, it is advantageous to constrain the kernel strongly and estimate only a low-dimensional parameterization of it.

In this thesis we do not attempt a fully blind deconvolution in the sense of estimating a completely free kernel together with a completely free latent image. Instead, for Problem **P1**, we exploit the known field layout and solve for the segment heights by least squares. For Problem **P2**, we again exploit the known field layout, but now allow one scalar kernel parameter to vary and estimate it together with the segment heights, so this falls under a semi-blind deconvolution method. In other words, the present work should be viewed neither as a generic image-deblurring method nor as a fully blind deconvolution method, but rather as a structured inverse problem approach that combines physical forward modeling with strong prior structural information about the wafer topography.

So, overall, the literature supports the methodological choices made in this thesis. In the non-blind deconvolution literature, least-squares type methods are very common and natural when the forward

Reference	Image prior	Kernel prior	Estimation strategy / remarks
Fergus et al. (2006)	Mixture of Gaussians on gradients (heavy-tailed natural image prior).	Sparsity prior, encourages zero-values, non-negative kernel (mixture of D exponential distributions).	Joint Bayesian estimation with variational inference, coarse-to-fine, alternating updates for (x, k).
Shan et al. (2008)	Sparse gradient prior + additional terms to suppress ringing / artifacts.	Regularized kernel with smoothness and support constraints.	Unified probabilistic model; alternating optimization of image and kernel with penalties to avoid trivial solutions.
Cho & Lee (2009)	Computationally efficient Gaussian prior.	Simple constraints (support, non-negativity).	Fast MAP-style method, prediction step predicts strong edges used for kernel estimation.
Krishnan & Fergus (2009)	Hyper-Laplacian prior (heavy-tailed).	Typically fixed (non-blind) or simple constraints in blind variants.	Efficient non-blind deconvolution (Lookup Table approach (LUT)); influential for showing effectiveness of heavy-tailed gradient priors for sharp, edge-preserving reconstructions.
Levin et al. (2009, 2011)	Sparse gradient prior. Sum over pixel gradients to the power α.	Uniform over feasible kernels (support, non-negativity, normalization).	Theory + experiments: naive $\text{MAP}_{x,k}$ biased toward δ-kernel; recommend marginal likelihood / MAP_k (optimize over k by integrating out x) plus separate non-blind deconvolution.
Deep learning methods (Nah et al. 2017, Kupyn et al. 2018)	Implicit learned prior via CNN/GAN; no explicit parametric image prior.	Blur implicitly modeled or jointly learned; sometimes no explicit kernel prior.	End-to-end training on synthetic/real pairs, strong performance but less interpretable, needs lots of data, not tailored to wafers specifically.

Table 2.1: Overview of choices for priors and estimation strategies for different papers.

model is fixed. The blind-deconvolution literature, especially Levin et al., explains why fully free joint estimation is not a good idea and why kernel uncertainty should be handled with caution. The faster optimization based papers show that exploiting edges and problem structure can make a big difference computationally. Finally, the deep-learning literature shows that strong results are possible without explicit kernel estimation, but also that such methods come with a loss of interpretability and with a heavy dependence on training data. In light of all of this, the structured least-squares and semi-blind formulations of this thesis are a natural and well-motivated choice.

3 | Least-squares algorithm

In Section 2.1, we formulated the two inverse problems that we consider in this thesis. Problem **P1** is the inverse problem with a fixed forward model, and Problem **P2** is the semi-blind version in which part of the kernel is also treated as unknown. In this chapter, we now turn to the corresponding reconstruction algorithms.

The main idea is that because the field layout is known a priori, the unknown wafer height profile can be parameterized by a finite-dimensional vector of segment heights. As a consequence, the inverse problem becomes a parameter-estimation problem rather than a function-recovery problem in an infinite-dimensional space. This makes it natural to formulate the reconstruction as a least-squares problem.

We will first describe the least-squares algorithm for Problem **P1**. Thereafter, we describe how this algorithm can be extended to the semi-blind setting of Problem **P2** by means of variable projection. In Section 3.2, these methods are then validated on synthetic data to see if we can retrieve the true height profile with good accuracy and there we will discuss the results in detail.

3.1 | Description of the algorithm

We will now describe the reconstruction algorithm in detail. As before, we work first in one dimension, that is, for a fixed value of x and along the scan direction y. Physically, this corresponds to selecting one spot of the level sensor and looking at the measurements obtained by that spot during a stroke.

Our starting point in this section is to continue where we left off in Section 2.1. So, we assume the same setting as described in Section 2 and Section 2.1.

Problem P1

Now, recall the first inverse problem, Problem **P1**:

$$\textbf{(P1)} \qquad \min_{\eta \in \mathbb{R}^S} \; \|z_{\text{meas}} - \Phi\eta\|^2 ,$$

where:

$$\Phi = \mathcal{S}_n \left(\mathcal{F}_n \left(\mathcal{L}_n \left(I_{\text{seg}} \right) \right) \right),$$

and the only unknowns are the segment heights collected in the vector η.
The data z_{meas} is assumed to be generated according to the model (for the synthetic data we know this for a fact):

$$z_{\text{meas}} = \mathcal{S}_n \left(\mathcal{F}_n \left(\mathcal{L}_n \left(h \right) + \varepsilon_n \right) \right).$$

Here, $\mathcal{L}_n$ denotes the level sensor response, $\mathcal{F}_n$ denotes the driver filter, $\mathcal{S}_n$ denotes the final mapping to the actual measurement locations, and ε_n denotes noise. The operator $\mathcal{S}_n$ is included in order to treat the synthetic-data and real-data case in one common framework: for synthetic data, $\mathcal{S}_n$ is simply the identity, whereas for real data it corresponds to the interpolation step from the internal 0.2 mm simulation grid to the actual measured within-field coordinates on a 0.5 mm grid.

How the matrix Φ is built in practice

Although we write:

$$\Phi = \mathcal{S}_n \left(\mathcal{F}_n \left(\mathcal{L}_n \left(I_{\text{seg}} \right) \right) \right),$$

in practice the matrix Φ is built column by column. This is done in a MATLAB implementation.

For each segment $j = 1, \ldots, S$, we take the j-th column of I_{seg}, which is the indicator vector of segment j, and apply to it:

1. the level sensor response $\mathcal{L}_n$ by using the simulation tool,

2. the driver filter $\mathcal{F}_n$ by using the simulation tool,
3. and, if necessary, the interpolation/sampling operator $\mathcal{S}_n$.

The resulting vector is the j-th column of Φ. Hence, if we denote the j-th column of Φ by $\phi_j \in \mathbb{R}^n$, then:

$$\phi_j = \mathcal{S}_n \left(\mathcal{F}_n \left(\mathcal{L}_n \left([I_{\text{seg}}]_{:,j} \right) \right) \right).$$

Stacking these columns yields Φ:

$$\Phi = [\phi_1, \ldots, \phi_S].$$

In this way, every column of Φ describes how that specific segment contributes to the final measured signal after passing through the full forward model. This specific construction is used both in the synthetic data setting and in the real-data setting.

Synthetic-data and real-data case

As we discussed, the algorithm is the same for the synthetic and the real data, except for one practical difference. For synthetic data, the measurements are generated directly on the internal 0.2 mm grid. Therefore, in that case we have:

$$\mathcal{S}_n = I_n,$$

with I the identity matrix. Hence the matrix Φ becomes:

$$\Phi = \mathcal{F}_n \left(\mathcal{L}_n \left(I_{\text{seg}} \right) \right).$$

For the real data, however, the measurements for a single field are available only on a much coarser 0.5 mm grid, which corresponds to about 65 or 66 within-field points. Since the ASML simulation tool still gives its output on the internal 0.2 mm grid, an interpolation step is necessary. So, in the real-data case, each simulated column ϕ_j, $j = 1, \ldots, S$, is first computed on the internal grid and is then interpolated to the actual measured within-field coordinates on the 0.5 mm grid. This interpolation step is exactly the operator $\mathcal{S}_n$ in the mathematical formulation.

Solving the least-squares problem

Once the matrix Φ has been constructed, we solve the least-squares problem (2.4):

$$\min_{\eta \in \mathbb{R}^S} \left\| z_{\text{meas}} - \Phi \eta \right\|^2.$$

In principle, if $\Phi^T \Phi$ is invertible, this problem has the analytical solution

$$\hat{\eta} = \left(\Phi^T \Phi \right)^{-1} \Phi^T z_{\text{meas}}.$$

However, in the MATLAB implementation used in this thesis, the solution is obtained using the MATLAB function `lsqr`. The reason for this is that `lsqr` is numerically more robust than explicitly forming and inverting $\Phi^T \Phi$, and it directly solves the least-squares problem that we are interested in. So, in practice, we compute:

$$\hat{\eta} = \arg\min_{\eta \in \mathbb{R}^S} \left\| z_{\text{meas}} - \Phi \eta \right\|^2$$

by applying `lsqr` to the system (Φ, z_{meas}). After the vector $\hat{\eta}$ has been found, the estimated/reconstructed sharp profile is obtained by:

$$h_{\text{est}} = I_{\text{seg}} \, \hat{\eta}. \tag{3.1}$$

This is the reconstructed step profile on the internal 0.2 mm grid.

Evaluation in measurement space

It is important to distinguish between the reconstructed/estimated step profile and the corresponding evaluated fit in the measurement space. Namely, what the least-squares method does is fit the output of the forward model $\Phi\eta$ to the measurements. This means that the least-squares method does not fit the step profile directly to the measurements. Intuitively, we can think of $\hat{\eta}$ as the vector of heights for the segments that best explains the measurements that we see, because if we apply to it the forward model, the result comes the closest to the measurements in the least-squares error sense.

Once $\hat{\eta}$ has been found, we compute:

$$z_{\text{fit}} = \Phi\hat{\eta}. \tag{3.2}$$

This quantity is the least-squares result of the algorithm fitting the output of the forward model to the measurements z_{meas}. In other words, z_{fit} is obtained by taking the reconstructed step profile and applying the same forward model to it. This allows us to compare the result to the measured data z_{meas}.

This distinction is important throughout the thesis. The vector:

$$h_{\text{est}} = I_{\text{seg}}\,\hat{\eta}$$

lives in the step profile space, whereas:

$$z_{\text{fit}} = \Phi\hat{\eta}$$

lives in the measurement space. The least-squares problem minimizes the residual in the measurement space, not the difference between the reconstructed sharp profile and some nominal reference pattern. Very importantly, this is exactly why, especially in the real-data case, a slightly better measurement-space fit does not necessarily imply a reconstructed sharp profile that is closer to the nominal alternating two-level pattern.

The algorithmic summary/pseudocode for Problem P1

So, the complete algorithm/pseudocode for Problem **P1** can be summarized as follows:

1. Construct the internal 0.2 mm grid $y_1, \ldots, y_n$ in the scan direction.
2. Use the known step locations to construct the indicator matrix I_{seg}.
3. For each segment $j = 1, \ldots, S$, apply the forward model to the j-th indicator vector (j-th column of I_{seg}) in order to obtain the j-th column of Φ, denoted by ϕ_j.
4. If the data are real filtered measurements, interpolate each simulated response column of Φ to the actual measurement locations.
5. Solve the least-squares problem and obtain:

$$\hat{\eta} = \arg\min_{\eta \in \mathbb{R}^S} \|z_{\text{meas}} - \Phi\eta\|^2 .$$

6. Construct the reconstructed sharp profile using the least-squares minimizer $\hat{\eta}$:

$$h_{\text{est}} = I_{\text{seg}}\hat{\eta}.$$

7. Evaluate the reconstructed/estimated step profile in measurement space by computing:

$$z_{\text{fit}} = \Phi\hat{\eta}.$$

Semi-blind extension: Problem P2

We will now describe the extension of the least-squares algorithm to the semi-blind deconvolution setting of Problem **P2**, where part of the kernel is also treated as unknown. Now, recall Problem **P2** (2.6):

$$\textbf{(P2)} \qquad \min_{\eta \in \mathbb{R}^S,\, \tau \in \mathbb{R}} \left\| z_{\text{meas}} - \Phi(\tau)\eta \right\|^2 .$$

This problem is clearly no longer linear in all unknowns, since the kernel depends nonlinearly on τ. However, for fixed τ, the problem is still linear in η:

$$\min_{\eta \in \mathbb{R}^S} \left\| z_{\text{meas}} - \Phi(\tau)\eta \right\|^2 .$$

This is a problem we already can solve, since it is the same as Problem **P1** but with a certain fixed parameter in the kernel. This observation is exactly what makes the variable projection method natural here. This is an instance of what is known as a separable nonlinear least-squares problem. In such problems, part of the unknowns enter the model linearly, while the remaining unknowns enter the model nonlinearly. The variable projection method, introduced by Golub and Pereyra and discussed for example in [3, 4, 11], exploits exactly this structure by eliminating the linear variables through an inner least-squares problem. In the present setting, the height parameters η are the linear variables, whereas the kernel parameter τ is the nonlinear variable. The reason that this splitting is valid is that if we define:

$$F(\eta, \tau) = \left\| z_{\text{meas}} - \Phi(\tau)\eta \right\|^2 ,$$

then the original problem (2.6) can be written as:

$$\min_{\eta,\tau} F(\eta, \tau) = \min_{\tau} \min_{\eta} F(\eta, \tau).$$

For every fixed value of τ, we may first solve the linear inner least-squares problem

$$\min_{\eta} F(\eta, \tau).$$

This gives the best possible height vector $\hat{\eta}(\tau)$ for that specific value of τ:

$$\hat{\eta}(\tau) = \arg\min_{\eta \in \mathbb{R}^S} \left\| z_{\text{meas}} - \Phi(\tau)\eta \right\|^2 .$$

Substituting this optimal height vector $\hat{\eta}(\tau)$ into the objective gives the reduced single-variable objective which we denote by $J(\tau)$:

$$J(\tau) := F(\hat{\eta}(\tau), \tau) = \left\| z_{\text{meas}} - \Phi(\tau)\hat{\eta}(\tau) \right\|^2 .$$

Minimizing $J(\tau)$ over τ is therefore equivalent to solving the original semi-blind least-squares problem (2.6), provided the reduced objective is minimized over the same admissible values of τ. This reduction is also computationally attractive. Instead of treating η and τ as one larger nonlinear optimization problem, the linear variables η are solved by a standard fast linear least-squares method for each trial value of τ. This is simpler, likely faster and more stable than jointly optimizing over all variables nonlinearly. The remaining nonlinear optimization is only one-dimensional in τ. Optimizing $J(\tau)$ over τ yields:

$$\hat{\tau} = \arg\min_{\tau \in \mathbb{R}} J(\tau).$$

Then, the final estimated segment-height vector is given by

$$\eta^* = \hat{\eta}(\hat{\tau}).$$

Intuitively, one can think of this as the height parameters that yield the closest fit to the measurements z_{meas} when we also optimize over τ in a specified bounded interval. Finally, the corresponding reconstructed height profile is given by

$$h_{\text{est}} = I_{\text{seg}}\, \eta^*.$$

How variable projection is implemented in practice

In practice, the semi-blind algorithm is implemented as follows. First, we choose an interval of admissible values for τ. Then, for every trial value of τ:

1. the matrix $\Phi(\tau)$ is built column by column,
2. the inner least-squares problem is solved using `lsqr`, yielding $\hat{\boldsymbol{\eta}}(\tau)$,
3. the reduced single-variable objective

$$J(\tau) = \|z_{\text{meas}} - \Phi(\tau)\hat{\boldsymbol{\eta}}(\tau)\|^2$$

is computed.

Finally, the reduced single-variable objective $J(\tau)$ is minimized over τ using a one-dimensional optimization procedure. In the MATLAB implementation used in this thesis, this is done using `fminbnd`. This is a built-in MATLAB function that finds a local minimum for a single-variable function over a specified bounded interval.

This means that the semi-blind algorithm remains conceptually very close to the least-squares algorithm for **P1**. The only additional step is that the forward model is now rebuilt for different trial values of τ, and for each such value the optimal profile parameters are computed by least squares. So the least-squares problem remains the central computational ingredient even in the semi-blind setting of Problem **P2**.

Remarks on the interpretation of the algorithm

It is worth emphasizing that the present algorithm is highly structured. We do not try to estimate an arbitrary unknown signal on an arbitrary grid, nor do we try to estimate a completely free unknown kernel. Instead, we make use of two very strong forms of prior structure:

1. the step locations inside the field are assumed known,
2. and in the semi-blind setting the kernel is only allowed to vary through a low-dimensional parameterization.

Because of this, the method is much simpler than a generic blind-deconvolution method from the image processing literature. This is what makes the least-squares approach and the variable-projection extension effective in the present setting. In other words, the main strength of the algorithm is not that it solves the most general possible deconvolution problem, but rather that it exploits the known geometrical and physical structure of the wafer topography and the measurement process as much as possible. In the next section, we will validate the algorithms that we described for Problem **P1** and Problem **P2** and discuss the results.

3.2 | Synthetic data: validation

Naturally, before applying the algorithms from Section 3.1 to real measurement data, it is important to first validate them in a setting where the true height profile is known exactly. This is precisely the role of the synthetic-data experiments in this section. By generating the data ourselves using the same simulation tool that is also used in the forward model, we know the true height profile, we know the true kernel settings, and we can therefore verify whether the reconstruction method is able to retrieve these quantities correctly.

The synthetic experiments serve two main purposes. Firstly, they provide a sanity check for the fixed forward model least-squares method for Problem **P1**. Secondly, they provide a controlled environment in which the semi-blind extension for Problem **P2** can be tested. This is very important, since in that case we also know the true value of the kernel parameter τ that was used to generate the data, whereas for the real data, we do not have that knowledge of the true value. Therefore it would be nice to verify if the algorithm can indeed retrieve both an accurate reconstructed step profile together with a value for τ that is close to the true value, before we apply it to real data.

Data generation process for synthetic data

Since ASML's simulation tool for the operators $\mathcal{L}_n$ and $\mathcal{F}_n$ works on a 0.2 mm grid, it is natural and straightforward to generate synthetic data on this same grid. This avoids any interpolation issues and allows us to first test the reconstruction method in the simplest possible setting.

We follow the exact steps as described in Section 2.1.1 to obtain (where we have $n = 166$ and $S = 17$):

- $y_1, \ldots, y_n$,
- $I_{\text{seg}} \in \mathbb{R}^{n \times S}$,
- $\eta_{\text{true}} = [\eta_1, \eta_2, \eta_1, \ldots, \eta_2, \eta_1]^T \in \mathbb{R}^S$, with $\eta_1 = 0$ nm and $\eta_2 = 150$ nm,
- $h_{\text{true}} = I_{\text{seg}}\, \eta_{\text{true}}, \quad h_{\text{true}} \in \mathbb{R}^n$.

The profile h_{true} is exactly the ideal step profile that we would like to recover from the synthetic measurements. The top left of Figure 3.1 shows a plot of h_{true}.

Next, the true profile h_{true} is passed through the forward model. In the fixed forward model case, we generate the synthetic measurements according to:

$$z_{\text{synth}} = \mathcal{F}_n\left(\mathcal{L}_n\left(h_{\text{true}}\right) + \varepsilon_n\right), \qquad \varepsilon \sim \mathcal{N}_n(0, \sigma^2 I),$$

where:

$$\sigma = 4.72 \text{ nm}.$$

So, the synthetic data is generated by first applying the level sensor response $\mathcal{L}_n$ to h_{true}. This action is plotted in the top right of Figure 3.1. Thereafter, we add the realistic Gaussian noise ε_n. In the bottom left plot of Figure 3.1, the result of adding this noise can be seen. Finally, we apply the filter $\mathcal{F}_n$ to obtain the final synthetic measurement signal z_{synth}, which can be seen in the bottom right plot of Figure 3.1. Since this all takes place on the internal 0.2 mm grid, we also have:

$$z_{\text{synth}} \in \mathbb{R}^n.$$

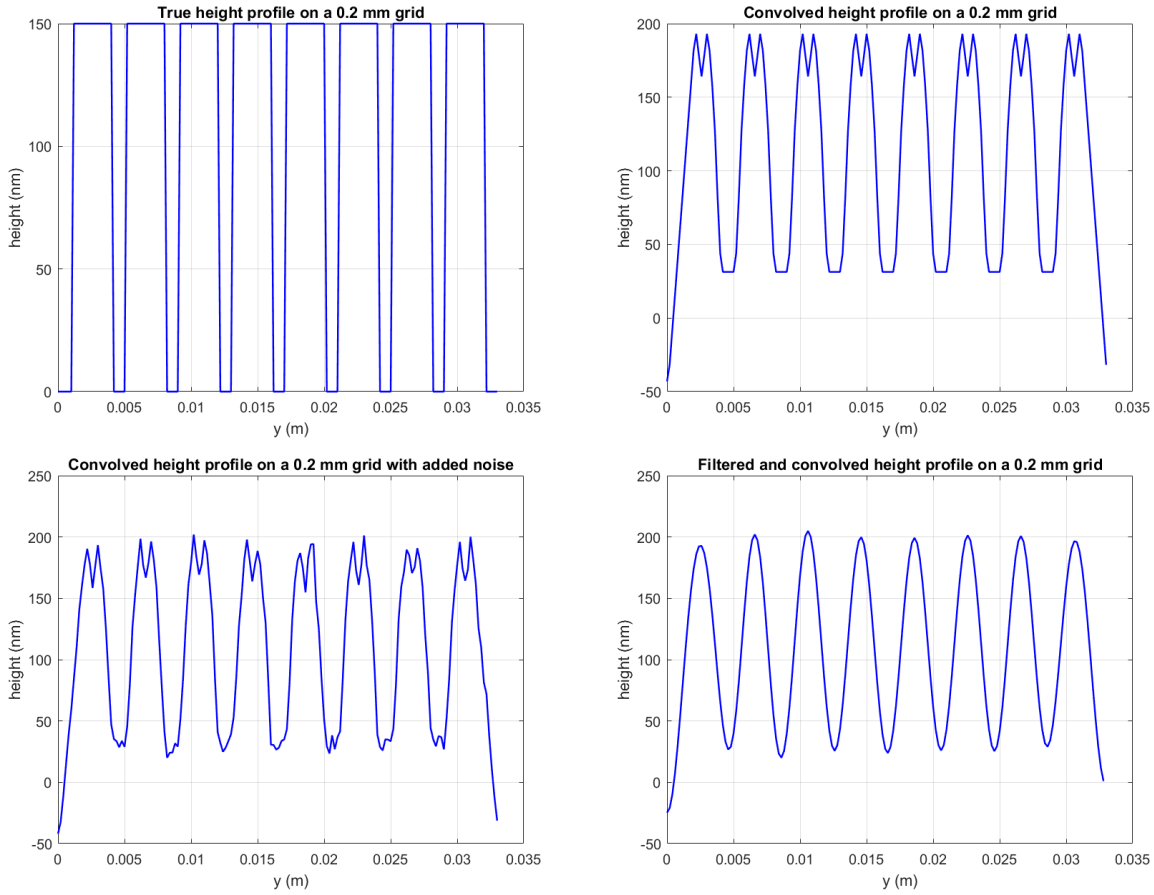


Figure 3.1: Synthetic data generation process. Top left: constructed height profile $\boldsymbol{h}_{\text{true}}$. Top right: convolved height profile $\mathcal{L}_n(\boldsymbol{h}_{\text{true}})$. Bottom left: convolved height profile with added noise. Bottom right: filtered and convolved (with added noise) height profile.

Visual structure of the synthetic signal

Before turning to the reconstruction results, it is useful to first briefly discuss what can be seen in the intermediate signals generated from the true step profile. The true height profile itself is of course piecewise constant and alternates between the low level and the high level. After applying the level sensor response $\mathcal{L}_n$, the profile is visibly smoothed. In particular, the trenches are pulled up by the neighbouring high segments, and the transitions are no longer sharp. In addition, around the height steps, clear artefacts caused by the curved wafer effect become visible. Adding the Gaussian noise also visibly changes the signal.

Finally, applying the filter $\mathcal{F}_n$ changes the signal again and yields the filtered measurement-space signal. In particular, applying the filter smooths the signal even more and the CWE artefacts are filtered out. By the time the final synthetic measurement vector $\boldsymbol{z}_{\text{synth}}$ is obtained, the sharp step profile has visibly already been transformed quite significantly. This also illustrates very clearly why the problem is an inverse problem: the measurements do not resemble the original height profile directly, but only a convolved, filtered and noisy version of it.

Validation of the least-squares method for Problem P1

We now apply the least-squares algorithm from Section 3.1 to the synthetic data. After solving the least-squares problem, we obtain the reconstructed height profile:

$$\boldsymbol{h}_{\text{est}} = I_{\text{seg}}\,\hat{\boldsymbol{\eta}},$$

and the corresponding evaluated fit in measurement space:

$$\boldsymbol{z}_{\text{fit}} = \Phi\hat{\boldsymbol{\eta}}.$$

In the left plot of Figure 3.2, the synthetic measurements are shown together with the true height profile h_{true} and the reconstructed height profile h_{est}. As can be seen, the reconstructed step profile comes extremely close to the true height profile and overlaps almost exactly in most places. So, from a first visual inspection, we can already conclude that the method seems to perform very well in this synthetic setting. In the right plot of Figure 3.2, the evaluated fit z_{fit} is shown together with the synthetic measurements z_{synth}. This is the quantity that is actually minimized in the least-squares problem by the algorithm. Again, the agreement is very close. Hence, both in sharp step profile space and in measurement space, the reconstruction appears to be highly accurate.

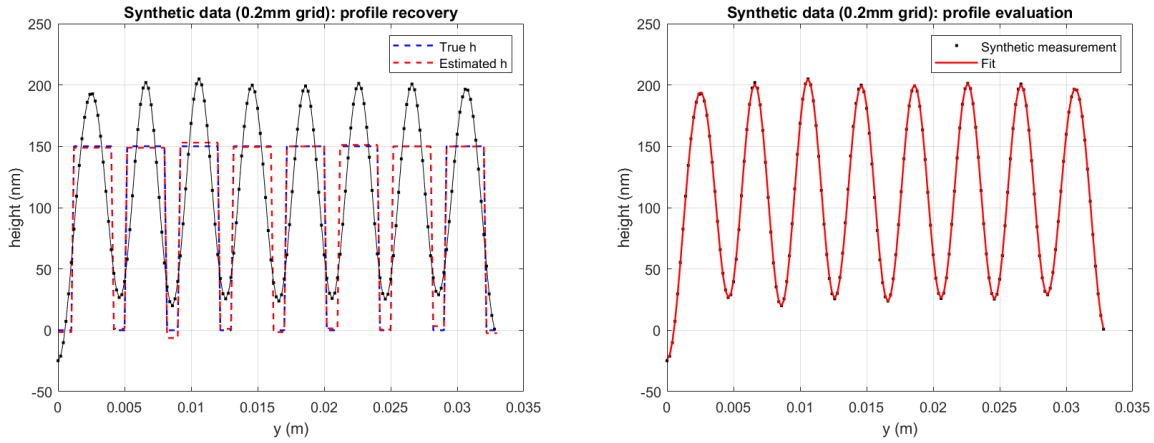


Figure 3.2: (Left) Comparison of synthetic measurements, reconstructed profile and true profile. (Right) Comparison of synthetic measurements and evaluated fit.

Quantitative error metric and analysis of impact of number of data points

In addition to a visual comparison, it is of course extremely useful to quantify the reconstruction performance. Since in the synthetic setting the true segment-height vector η_{true} is known exactly, multiple error metrics can be computed. The main metric that we use throughout this thesis is what we will call the offset-invariant relative error:

$$e_{\text{off-inv}} = \frac{\left\| \left(\hat{\eta} - \bar{\hat{\eta}}\,\mathbf{1}\right) - \left(\eta_{\text{true}} - \bar{\eta}_{\text{true}}\,\mathbf{1}\right) \right\|}{\left\| \eta_{\text{true}} - \bar{\eta}_{\text{true}}\,\mathbf{1} \right\|}, \tag{3.3}$$

where $\mathbf{1}$ is a vector of ones in $\mathbb{R}^S$. This metric compares the recovered segment-height vector $\hat{\eta}$ with the true (or in the case of real data, nominal) alternating two-level pattern η_{true} after removing the mean from both vectors. By doing this, global height offsets do not influence the metric: if the recovered height vector is equal to the nominal height vector plus the same constant added to each entry (so the nominal height vector but shifted up or down with a constant), the error will still be zero. This is very useful, because in later chapters we will also want to compare synthetic and real-data results in a way that is not influenced by an absolute offset. This will also allow use to compare the accuracy for the real data setting with the accuracy in the synthetic setting.

In addition, this metric is stronger and more appropriate than a pure contrast/step based metric. That would only check whether the average difference between the recovered low and high segments is close to 150 nm. The offset-invariant relative error, however, evaluates the full recovered vector and therefore also penalizes deviations among segments that should in principle belong to the same level. In other words, it quantifies how much the recovered height vector looks like the true/nominal height vector while ignoring any global offset that might occur.

For Problem **P1** on the synthetic data, we obtain:

$$e_{\text{off-inv}} = 2.891 \cdot 10^{-2}.$$

So, the recovered segment-height vector differs by only about 2.89% from the true alternating two-level nominal pattern. This is a reasonably small error and indicates that the recovered height vector is very close to the true one in the synthetic setting. Hence, this metric quantitatively confirms what

is already visible in the plots: namely that the least-squares method is able to retrieve the true step profile with with very high accuracy in the synthetic setting. The synthetic validation shows that the formulation of Problem **P1** and the associated least-squares algorithm are correct. This is important, because it means that if the forward model is known and the data is generated according to that same forward model, then the reconstruction method is able to retrieve the original segment heights very accurately.

Additionally, to study the effect of the number of measurement points on the reconstruction error, we repeated the synthetic reconstruction while using only m measurement locations. The forward model was still generated on the 0.2 mm grid, but the final filtered signal and the corresponding columns of Φ were evaluated only at m uniformly spaced locations. As can be seen in Figure 3.3, the resulting error $e_{\text{off-inv}}$ initially decreases as m increases, but after a certain number of measurement points the curve levels off. We suspect that this is caused by the fact that the driver filter also acts on the noise. Therefore, the final measurement noise is no longer independent at each measurement location, but becomes filtered and correlated in the scan direction. As a result, increasing m beyond a certain point might not add much independent new information to the least-squares problem.

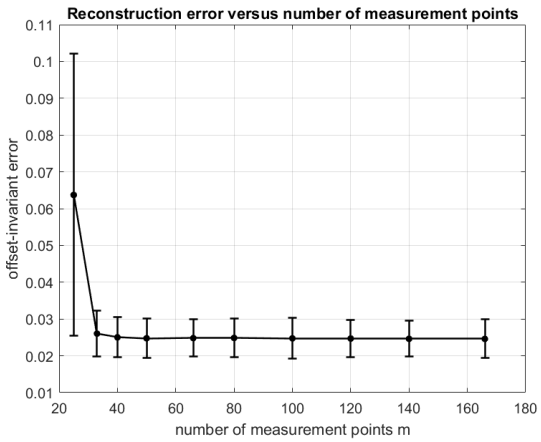


Figure 3.3: Reconstruction error $e_{\text{off-inv}}$ vs number of measurement points m. The error bars indicate one empirical standard deviation over 1000 independent noise realizations for each value of m.

Validation of the semi-blind method for Problem P2

We now turn to the semi-blind setting of Problem **P2** and apply the second algorithm from Section 3.1 to synthetic data. Since we are in the semi-blind synthetic experiment, we not only choose a true height profile, but also a true value of the kernel parameter τ_{true}. For this experiment we choose $\tau_{\text{true}} = 0.3$ mm. The synthetic measurements are then generated according to:

$$z_{\text{synth}} = \mathcal{F}_n\left(\mathcal{L}_{n,\tau_{\text{true}}}\left(h_{\text{true}}\right) + \varepsilon_n\right).$$

So, in this synthetic setting, both the true segment-height vector η_{true} and the true kernel parameter τ_{true} are known. We then apply the semi-blind reconstruction method described in Section 3.1. An important plot in this setting is the graph of $J(\tau)$ as a function of τ, which can be seen in Figure 3.4.
If we add no noise at all, this reduced objective attains its minimum exactly at the true value τ_{true}. This is an important sanity check, since it shows that the variable-projection implementation is fully consistent with the forward model used to generate the data. When realistic Gaussian noise is added (the same noise as for Problem **P1**), the minimizing value $\hat{\tau}$ is generally no longer exactly equal to τ_{true}. This is expected: we are then minimizing the residual for one specific realization of the random noise. Nevertheless, the estimated $\hat{\tau}$ remains very close to the true value. In particular, we can compute the relative error in τ as:

$$e_{\tau\text{-rel}} = \frac{|\hat{\tau} - \tau_{\text{true}}|}{|\tau_{\text{true}}|}.$$

Note that this quantity cannot be computed for the real data, since the true value of τ is unknown there. After 7 iterations of the optimizer, we find:

$$\hat{\tau} = 0.3074 \text{ mm}, \quad e_{\tau\text{-rel}} = 2.472 \cdot 10^{-2}.$$

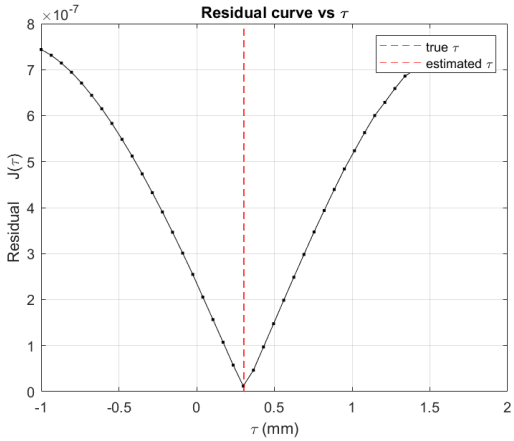


Figure 3.4: Reduced single-variable objective $J(\tau)$ vs τ.

So our estimate of τ only differs 2.47% from the true value and the reduced objective $J(\tau)$ still has a clear minimum near τ_{true}. This indicates that the parameter τ is identifiable from the data in this setting. Figure 3.5 shows two plots pertaining to the convergence of the optimizer. In the left plot, the 'best-so-far' value of $J(\tau)$ is shown. In the right plot, the 'best-so-far' value of τ is shown.

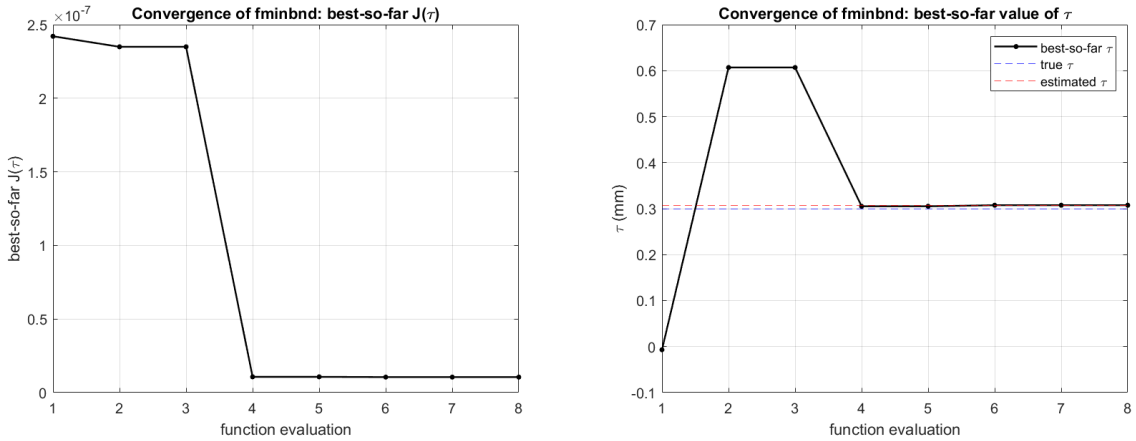


Figure 3.5: (Left) 'Best-so-far' value of $J(\tau)$. (Right) 'Best-so-far' value of τ.

After substituting the estimated value $\hat{\tau}$ into the forward model, the corresponding inner least-squares problem yields the reconstructed segment-height vector $\hat{\eta}(\hat{\tau})$, and hence the reconstructed height profile

$$h_{\text{est}} = I_{\text{seg}}\, \hat{\eta}(\hat{\tau}) = I_{\text{seg}}\, \eta^*.$$

This estimated step profile is plotted together with the true step profile and the synthetic measurements in Figure 3.6. As we can see, the estimated profile again comes very close to the true step profile, while the evaluated fit

$$\Phi(\hat{\tau})\hat{\eta}(\hat{\tau})$$

also matches the synthetic measurements very well.
For Problem **P2** on the synthetic data, we obtain:

$$e_{\text{off-inv}} = 2.999 \cdot 10^{-2}.$$

So, using the estimated parameter $\hat{\tau}$, the recovered segment-height vector differs by only about 3.00% from the true alternating two-level nominal pattern. Even though this error is slightly larger than that of Problem **P1**, this is still a reasonably small error, and it makes sense for it to be slightly worse than that for Problem **P2** due to the increased complexity of Problem **P2**. So, the recovered height vector is still

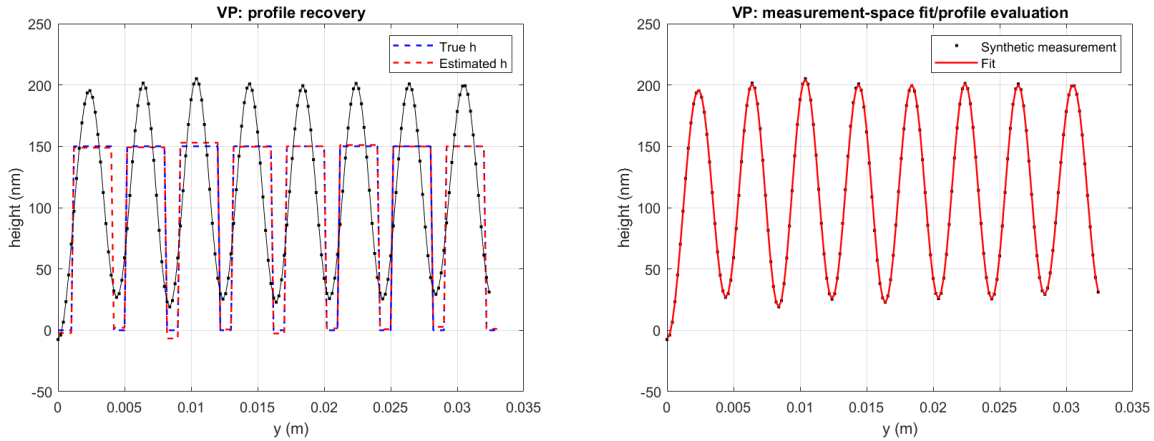


Figure 3.6: (Left) Comparison of synthetic measurements, reconstructed profile (for estimated τ) and true profile. (Right) Comparison of synthetic measurements and evaluated fit.

very close to the true one in the synthetic semi-blind deconvolution setting. This metric quantitatively confirms what is already visible in the plots: namely that our semi-blind deconvolution method is able to retrieve the true step profile *and* the true kernel parameter with very high accuracy in the synthetic setting.

From the synthetic experiments, we may conclude that the mathematical formulations of the inverse problems are sound, that the numerical implementations are correct, and that both the fixed forward model and semi-blind methods are sufficiently validated to be applied to real measurement data in the next chapter.

4 | Results for real data

Now, we can apply our methods to real measurement data instead of synthetic data. We will be looking at real, filtered data that has been measured with two different measurement strategies. We shall call these two measurement strategies 'A' and 'B' and leave the proprietary details out of this thesis.

It is important to realize that for the simulated signal, we worked with 166 measurements over the 33 mm of a single field, whereas the real data only contains 65 or 66 measurements for a single field, depending on the measurement strategy used. We already expect this lesser degree of information to impact the results.

We make sure to select a spot based on its x-location such that we get the desired cross section of the field in the y-direction as is the case in the right plot in Figure 2.1. We naturally want to avoid the trivial case where the spot has an x-value right in the center of the field causing it to measure a constant height instead of the characteristic step profile we are interested in analyzing.

Removing the global shape of the wafer

Wafers have a so-called *global shape*. We can think of this as a broad, low-frequent curvature or bend in base of the wafer itself. As one might expect, this impacts the measured height profile quite significantly. The plot on the left in Figure 4.1 shows height measurements of a full stroke for a single spot. This is real filtered data. The global shape of the wafer is very visible here.

Ideally, we want the wafer itself to be perfectly flat, so that we only measure and can thereafter purely analyze the actual topography of the fields. Fortunately, we can for a big part remove this global shape by subtracting a polynomial fit from the wafermap. Removing such a polynomial fit yields the plot on the right in Figure 4.1. Clearly, the impact of the global shape has been reduced quite significantly and we can see the recurring field pattern on more of a flat line now. Of course, there is still some global shape/recurring bending visible. We also expect that this might very well impact the results, since for a single field, there will still be some bending taking place.

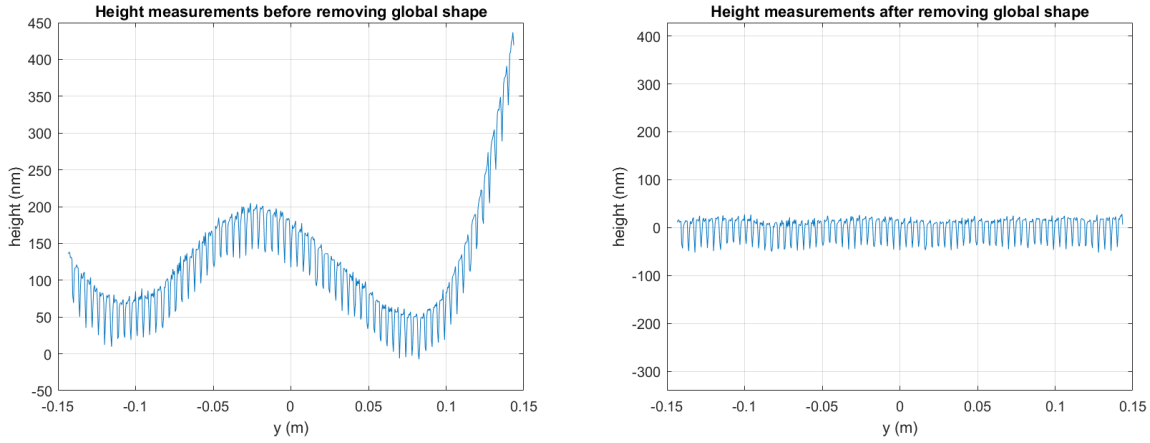


Figure 4.1: Height measurements of a full stroke, before and after removing the global shape of the wafer.

4.1 | Results on real 1D data

Before we get into the results, an important fact to keep in mind throughout this section is that, unlike in the synthetic setting, we do not know the true step profile of the real field exactly. In the figures below, the blue 'nominal' profile is not a completely true profile that we know to be true. Instead, it is the nominal alternating two-level profile with step height 150 nm (even this 150 nm we cannot be completely sure of), centered around the center of the estimated segment height vector. In other words, the nominal profile is shifted vertically such that it has the same mean as the estimated profile. This has been done purely to give visual intuition and a visual benchmark to the reader. Without such a centering step, the

comparison would be much harder to interpret visually, since the absolute vertical offset of the real data is not known exactly.

Problem P1: fixed forward model - Measurement Strategy A

We first consider Problem **P1**, the least-squares reconstruction with the fixed forward model. In this case, we use the same kernel settings as in the synthetic experiments and set the kernel parameter equal to:

$$\tau = 0.0 \text{ mm}.$$

Applying the reconstruction method to the selected field yields the results shown in Figure 4.2. In the left plot, the centered nominal profile, the estimated sharp profile and the real measurements are shown together. In the plot on the right, the evaluated measurement-space fit is shown together with the real measurements.

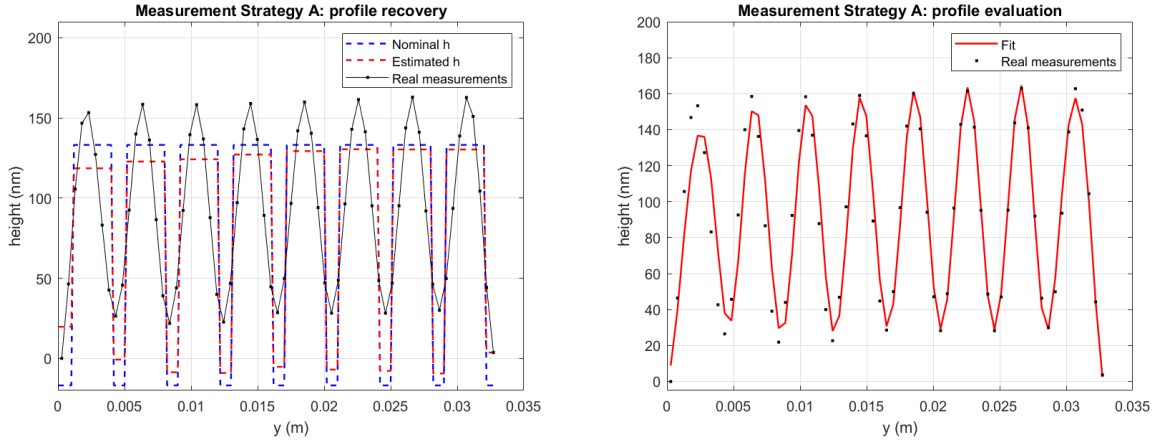


Figure 4.2: (Left) Comparison of real measurements, reconstructed profile and nominal profile. (Right) Comparison of real measurements and evaluated fit.

A first observation is that the result is clearly worse than in the synthetic setting, which is of course not surprising. The recovered profile still follows the nominal alternating structure reasonably well, but the nominal and estimated profiles are visibly less close than in the synthetic case. Likewise, the evaluated measurement-space fit is clearly not as accurate as for the synthetic data, although it still captures the overall structure of the measurements quite well. In particular, the right half of the field is actually fitted rather well, whereas the mismatch is more pronounced on the left half.

Quantitatively, for this real data reconstruction we obtain:

$$e_{\text{off-inv}} = 1.672 \cdot 10^{-1}.$$

So the offset-invariant relative error is about 16.7%. This is substantially larger than in the synthetic setting, where the same metric was only 2.89%. Still, considering that we are now working with real filtered measurements rather than synthetic data generated exactly from the forward model, this result is not poor at all. It indicates that the recovered segment height vector still resembles the nominal alternating two-level pattern reasonably well, but clearly with a noticeably larger deviation than in the ideal synthetic case. This is consistent with what is already visible in the plots: the evaluated fit in measurement space is decent, but clearly less accurate than for synthetic data.

There are several natural reasons for this decrease in performance. First of all, because we are applying the reconstruction algorithm on real data, the additional interpolation step is necessary, which could potentially impact the accuracy of our method. Moreover, the real data on a 0.5 mm grid is simply a lot sparser than the synthetic data generated on a 0.2 mm grid, so in the real-data case, we are guaranteed to have less information to go off. This certainly impacts the performance in a negative way.

Secondly, the global shape of the wafer first has to be removed. Although this is necessary, it is of course only an approximation. If the global shape removal is not perfect, then residual trends may remain in the

data of the local field and part of the structure may be slightly distorted. This can negatively affect both the recovered step profile and the evaluated measurement-space fit.

Thirdly, and very importantly, there is almost certainly model mismatch; in the synthetic setting, the data is generated by exactly the same forward model as the one used in the reconstruction. In the real data setting, this is no longer true. The true physical measurement process may differ from the model in multiple ways: the true kernels may significantly differ from the modeled kernels. Additionally, the chosen parameter settings (for example for the CWE) may be inaccurate, the true value of the τ parameter is unknown, the filtering may not be modeled perfectly, and the actual field may deviate from the nominal profile. In addition, the real data most likely contains effects that are not explicitly included in the present formulation, such as residual measurement artefacts from the sensor and other machine specific artefacts.

So, taken together, the larger value of $e_{\text{off-inv}}$ and a larger residual are exactly what one would expect when moving from an ideal synthetic setting to real filtered measurement data. The important point is that the method still recovers the correct qualitative step structure with values relatively close to what we know to be true, and yields a fit that is clearly still meaningful.

Problem P2: semi-blind deconvolution with unknown kernel parameter - Measurement Strategy A

We now consider Problem **P2**, in which the kernel parameter τ is treated as unknown and estimated from the real data together with the segment height vector. It is important to emphasize that we are using the same field with the same measurements as for Problem **P1**. Additionally, note that we do not know the true value of τ since we are working with real data. In this case, the reduced objective

$$J(\tau) = \|z_{\text{meas}} - \Phi(\tau)\hat{\eta}(\tau)\|^2$$

is minimized over an interval of admissible values of τ: $[-1.5 \text{ mm}, 1.5 \text{ mm}]$. For the selected real dataset and field, after 21 iterations of the optimizer `fminbnd`, this procedure yields:

$$\hat{\tau} = 0.4 \text{ mm}.$$

This means that this value for τ gives the best measurement-space fit among the values considered and is the minimizer of $J(\tau)$. In Figure 4.3, the residuals can be seen for the different values of τ in the interval.

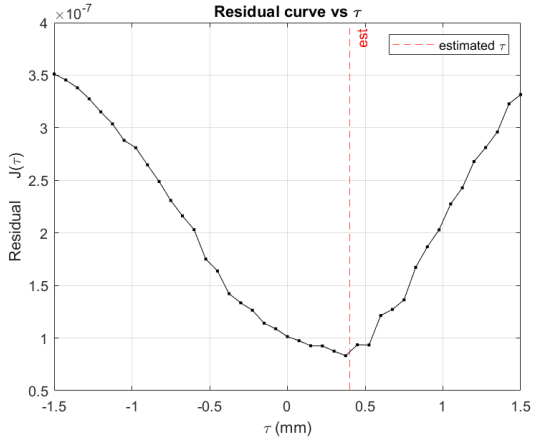


Figure 4.3: Reduced single-variable objective $J(\tau)$ vs τ.

At the same time, for this value of τ, the offset-invariant error becomes:

$$e_{\text{off-inv}} = 2.190 \cdot 10^{-1},$$

which translates to 21.9%, which is actually worse than what we found for Problem **P1**. So the semi-blind method improves the measurement-space fit, but yields a recovered step profile that is further away from the nominal pattern according to the offset-invariant metric. At first sight, this may seem counterintuitive, but it is in fact a very natural consequence of the structure of the inverse problem. The key point is that Problem **P2** is formulated to optimize the measurement-space residual, not the closeness of the recovered

step profile to the nominal pattern. In other words, the semi-blind method chooses the value of τ that makes

$$\Phi(\tau)\hat{\eta}(\tau)$$

fit the real measured signal as well as possible. It does not explicitly try to make the vector $\hat{\eta}(\tau)$ look as much as possible like the nominal vector

$$[0, 150, 0, 150, \ldots]^T \in \mathbb{R}^S.$$

This is a very important detail. The forward model is smoothing and filtering. So it maps a sharp step profile to a much smoother measured signal. Because of this, different sharp step profiles can lead to rather similar measurement-space signals after the forward model has been applied. Hence, it is entirely possible that one specific value of τ leads to a recovered profile that looks less like the nominal profile, but after convolution and filtering explains the measurements better. This effect is especially plausible in the real data case, since here we do not know the true value of τ and we do not know the true step profile exactly either. So the fact that $\hat{\tau} = 0.4$ mm improves the fit does not necessarily mean that this is the physically true τ value for this dataset. It only means that, within the present model class, this is the value that yields the best agreement with the measured data. Additionally, because there is some degree of model mismatch, the parameter τ can also partly compensate for imperfections elsewhere in the forward model. As a result, the optimal measurement-space parameter does not need to coincide with the parameter that yields the most nominal looking step profile.

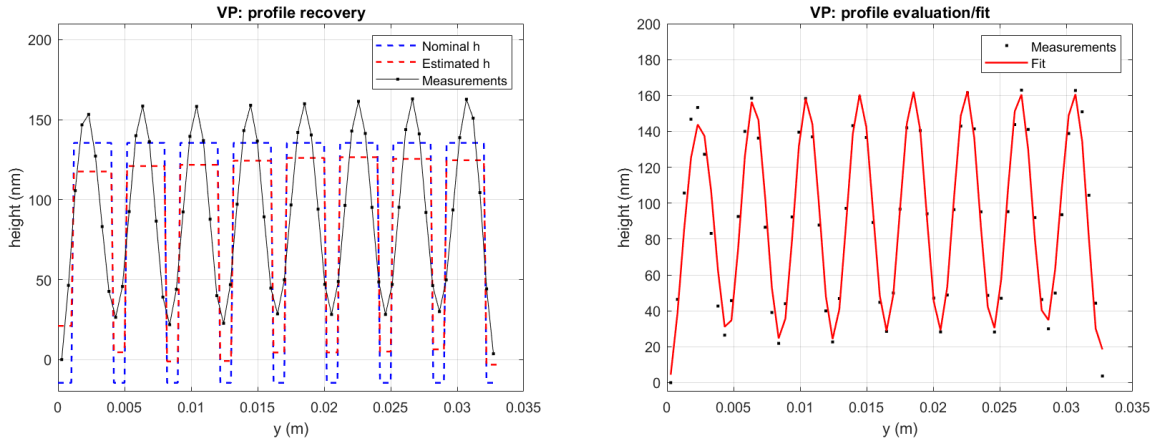


Figure 4.4: (Left) Comparison of real measurements, reconstructed profile (for estimated τ) and nominal profile. (Right) Comparison of real measurements and evaluated fit.

4.2 | Extension of the 1D problem to multiple spots

We now extend the one-dimensional real-data reconstruction from the previous section to a two-dimensional map of the same field and the same dataset. As already mentioned in the introduction, the idea here is very straightforward: instead of selecting only one spot of the level sensor and reconstructing the height profile in the scan direction for that one spot, we now select all spots in the stroke and apply exactly the same real-data reconstruction method from Problem **P1** to each of them separately.

So, for every spot position in the x-direction, we again solve the one-dimensional least-squares problem for the unknown segment-height vector. In this way, each spot yields its own recovered step profile in the y-direction. By placing these one-dimensional reconstructions next to each other at their corresponding x-locations, a two-dimensional map of the field is obtained.

It is important to emphasize once more that this does *not* constitute a full two-dimensional deconvolution problem in both x and y. We do not perform any deconvolution in the x-direction, and we do not reconstruct a completely free two-dimensional surface. What we do instead is repeat the one-dimensional reconstruction independently for all spots and then combine the results into a spatial map at the discrete spot locations in the x-direction. So the obtained two-dimensional map should be interpreted exactly in this sense.

Recovered step profile

Multiple plots of the recovered two-dimensional step profile can be seen in Figure 4.5. In the top left plot, the recovered step profile can be seen at an angle from which the recurring trenches are visible. In the top right plot, the same recovered step profile can be seen from a similar angle but with the measured data overlayed in grey. The plot shows very clearly that the characteristic structure is recovered over the entire field and not only for a single spot; the recurring alternating step pattern in the scan direction is visible across essentially all spot positions. In the bottom left plot of Figure 4.5, the same recovered two-dimensional profile is shown from an alternate angle, and in the bottom right plot, we show a view from the very top of the field, so that it reads more like a heatmap. To generate these plots we used the function `surf` in MATLAB. These figures confirm that the one-dimensional reconstruction from the previous section is not an isolated result for one particular spot, but that the same type of recovered profile persists across the full spot array. Taking the average of the offset-invariant errors of all spots yields:

$$\bar{e}_{\text{off-inv}} = 1.731 \cdot 10^{-1}.$$

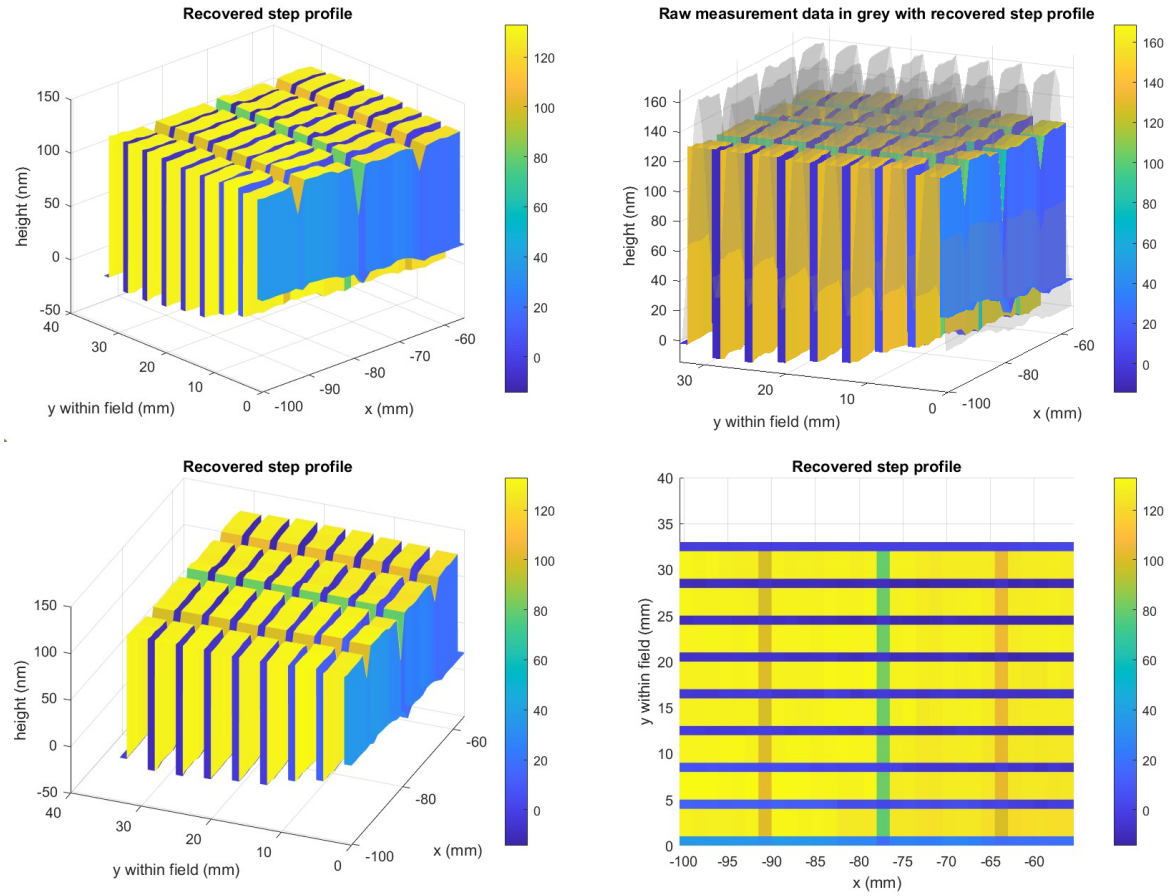


Figure 4.5: (Top left) Reconstruction of step profile for a single field. (Top right) Reconstruction for a single field with measured data overlayed. (Bottom left) Reconstruction of step profile for a single field from an alternative angle. (Bottom right) Reconstruction for a single field from a top view.

Concluding remarks

The conclusion of this two-dimensional extension is mainly that the one-dimensional reconstruction appears to work in a consistent way over all spots and gives a visually meaningful result when combined into one a single map for a selected field. Of course, this is not yet a full two-dimensional deconvolution problem in both x and y. Nevertheless, it is still a useful and satisfying result, since it shows the recurring step structure clearly over the full field and supports the idea that the reconstruction behaves consistently across the spot array. In that sense, the main value of this two-dimensional extension is that it gives an intuitive and informative visualization of the reconstructed field.

5 | Conclusion

5.1 | Summary of the main findings

In this thesis, we studied the inverse problem of recovering the original wafer height profile from level sensor measurements. More specifically, we considered wafers with known within-field topography and used this prior structural information to reduce the inverse problem to a finite-dimensional parameter-estimation problem. Instead of trying to recover an arbitrary function, we assumed a piecewise constant profile with known segment boundaries/step locations and solved for the unknown segment heights.

The first main finding of this thesis is that the least-squares formulation works very well in the synthetic setting. When synthetic data is generated on the internal 0.2 mm grid using the same forward model as the one used in the reconstruction, the method is able to recover the original alternating step profile with very high accuracy. Both visually and quantitatively, the reconstructed profile comes extremely close to the true profile. So, this validates the mathematical formulation of Problem **P1** and confirms that the implementation of the least-squares method is correct.

The second main finding is that the same least-squares framework can also be applied successfully to real filtered level sensor data. In the real data case, an additional interpolation step is necessary, since the simulation tools for the forward model operate on a 0.2 mm grid whereas the actual measurements for one field are available only on a coarser 0.5 mm grid. Even with this additional complication, the method is still able to recover the characteristic step structure of the field in a meaningful way. Naturally, the performance on real data is clearly worse than on synthetic data, which is exactly what one would expect due to sparser data, the additional interpolation step, global-shape removal, model mismatch and the fact that the real measurements are not generated by the forward model exactly. Still, the results show that the method remains useful in practice and does not only work in an idealized synthetic setting.

The third main finding is that a first semi-blind deconvolution experiment is feasible in the present setting. By introducing a scalar kernel parameter τ into the forward model and solving the resulting inverse problem by variable projection, we showed that it is possible to jointly estimate the height profile and part of the kernel. In the synthetic setting, this semi-blind method performs very well: the true value of τ is recovered accurately and the underlying step profile is reconstructed very well at the same time. So, this demonstrates that it is meaningful to move beyond a completely fixed forward model and to estimate part of the kernel from the data as well.

A fourth important finding follows from applying this semi-blind algorithm to real data. There, the value of τ that gives the best measurement-space fit is not necessarily the value that yields a recovered step profile that is closest to the nominal alternating pattern. This is an important observation, since it reflects the structure of the inverse problem itself. The method optimizes the fit after application of the forward model, that is, in measurement space. It does not directly optimize closeness to the nominal step profile. So, in the real-data case, improved measurement-space fit does not automatically imply a reconstructed sharp profile that looks more like the nominal profile. This distinction is one of the main conceptual insights of this thesis.

Finally, the one-dimensional real-data reconstruction can be extended naturally to a two-dimensional map of the selected field by applying the same one-dimensional method independently to all spots in a stroke. This does not yet constitute a full two-dimensional deconvolution problem in both spatial directions x and y, but it does yield a spatial map of the reconstructed field in the y-direction at the discrete spot locations in the x-direction. In this way, the reconstructed step structure can be visualized over an entire field rather than only for one single spot, and the consistency of the reconstruction across the array of spots becomes visible as well.

So, taken together, the main conclusion of this thesis is that a least-squares approach based on known field layout and piecewise constant parameterization of the height profile is a viable and effective way to reconstruct wafer topography from level sensor measurements. In the fixed forward model setting, the method performs extremely well on synthetic data and still meaningfully well on real data and a first semi-blind extension based on variable projection is feasible and gives promising results. Hence, the main research question of this thesis can be answered positively: given convolved, filtered and noisy level sensor measurements, it is indeed possible to retrieve the underlying height profile of the wafer to a meaningful

extent, provided that sufficient structural information about the field layout is incorporated into the model.

5.2 | Discussion and limitations

In this thesis, we formulated the deconvolution problem for the level sensor as an inverse problem and showed that this leads to a method that performs very well on synthetic data and still meaningfully well on real filtered data. At the same time however, it is important to discuss what the main limitations of the current approach are.

A first important limitation is model mismatch. In the real-data setting, the true physical measurement process is of course not exactly equal to that of the mathematical model that we use. The level sensor kernel, the driver filter, the true value of the kernel parameter τ, the precise noise structure and possible residual artefacts are not known perfectly. So even if the true height profile would lie inside the model class that we consider, there can still be a mismatch between the actual machine behavior and the forward model used in the reconstruction. This is also likely one of the main reasons why the real-data performance is clearly worse than the performance on synthetic data.

A second limitation is pertaining to the sparsity of the real data. The real data is only available on a 0.5 mm grid, yielding approximately 65 sampling points for an entire field. This is a lot sparser for the 166 points for the synthetic data, and this is likely also a big explaining factor behind the performance drop when we go from synthetic to real data, since the real data simply contains less information than the synthetic data. This makes it harder to distinguish between different step profiles that may lead to similar measurement-space results after applying the forward model. Moreover, since the simulation for $\mathcal{L}$ and $\mathcal{F}$ works on a 0.2 mm grid, there is an additional extrapolation step necessary, which might impact the accuracy.

A third limitation is again related to the real data. Namely that the global wafer shape is included in the measurements and that this first has to be removed before a field can be analyzed. This is necessary, but of course only approximate; the global shape removal is likely not perfect (often we can still see the trend occur in the data visually), which causes some amount of global shape still to be present in the data for a field. These residual global shape trends that remain are then partly absorbed by the estimated segment heights. So, part of the error in the reconstruction process for the real data may not come from the deconvolution algorithm itself, but already from preprocessing.

Another important limitation is the assumption that the step locations are fully known and fixed. This is a strong assumption and it is one of the reasons why the inverse problem becomes tractable in the present form. In the synthetic setting, this assumption is of course exactly correct by construction. In the real-data setting, however, the true step locations are not known with complete certainty. That said, based on the real-data results, the nominal step locations appear to check out rather well visually; the reconstructed profiles line up with the alternating structure of the data. Specifically, in the trench segments, the lowest measurement data point often falls quite neatly into the trench segment. We have not encountered a specific instance of real data for a field where these low measured points fall into a completely wrong neighboring segment, this can be seen in the left plot of Figure 4.2. However, the assumption of these step locations is quite strong, and optimizing over the step locations as well could improve the results, although we do not think this assumption is the dominant source of error.

A further limitation is that the present method uses a very specific model class, namely piecewise constant profiles with known segment boundaries. This is a strength and a limitation at the same time. It is a strength because it incorporates strong prior information and makes the inverse problem much more manageable. But it is also a limitation because any deviation of the real field from this idealized piecewise constant structure cannot be represented explicitly by the model. So, if the true field contains more subtle structure within a segment, the reconstruction will necessarily absorb this into the segment heights in an approximate way rather than represent it directly.

Finally, for Problem **P2**, an additional limitation is that only one scalar kernel parameter is allowed to vary. This makes the semi-blind problem well-structured and manageable, but it also means that only a very limited amount of model uncertainty is captured. Therefore in the real-data setting, the value of τ

that gives the best fit in measurement space should not be interpreted too literally as the physically true parameter value. Additionally, other effects can also be compensated for and absorbed by this τ. Rather, τ is better understood as the value that, within the present model class, compensates best for the mismatch between the measured data and the forward model. This is also exactly why a better measurement-space fit can come together with a recovered step profile that is actually less close to the nominal alternating pattern.

5.3 | Recommendations for future work

There are several directions for future work that follow from the results and limitations discussed above.

A first recommendation is to investigate the forward model in more detail in order to improve the accuracy of reconstruction based on real data. Perhaps the simulation models for $\mathcal{L}$ and $\mathcal{F}$ can be further extended and additionally made more accurate. In particular, it would be interesting to investigate whether introducing more parameters in the kernel $\mathcal{L}$ than only a single scalar parameter τ can yield better reconstructions.

This leads to a second natural direction for future work, which is to extend the semi-blind deconvolution approach further. In this thesis, Problem **P2** only allows one kernel parameter to vary, which keeps the problem tractable and makes variable projection a very natural method. A logical next step would be to investigate which additional parameters can be created for the kernel, and thereafter if two or more kernel parameters can be estimated simultaneously. This could make the forward model more flexible and possibly reduce model mismatch in the real-data case and thereby increasing accuracy. At the same time, this would also make the optimization problem more difficult and potentially less stable.

A third recommendation is to investigate the step locations themselves and include these in the optimization. Even though visually these step locations seem to be quite accurate, it could be interesting to investigate in future work whether additional optimization of the step locations improves the results, especially for real data.

Another very important recommendation is to find a way to make the real data denser. This is very likely to improve the accuracy of the method for the real data. We tried to make the real data denser by superposing fields on top of each other, but because the within-field coordinates were exactly the same for all of the fields, for multiple measurement strategies, this approach was not successful in making the real data any denser. We expect that if there is a way to the real data meaningfully denser by superposing the fields, or through another way, that this would drastically improve the accuracy of the results for the real data.

Finally, it would be very interesting to extend the presented results to a two-dimensional inverse deconvolution problem in both spatial directions. Such an extension would be significantly more complex, but it could also make more use of the available structure of the field.

6 | References

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